



NEET (UG)-2026

|| DATE: 03-05-2026 ||

Important Instructions:

1. The test is of **3 hours** duration and the Test Booklet contains **180** multiple-choice questions (four options with a single correct answer) from **Physics, Chemistry, and Biology (Botany and Zoology)**. All questions are compulsory.
2. Each question carries **4 marks**. For each correct response, the candidate will get **4 marks**. For each incorrect response, **1 mark will be deducted** from the total scores. The maximum mark is **720**.
3. Use a **Blue/Black** ballpoint Pen only for writing particulars on this page/markings responses on the Answer Sheet.
4. Rough work is to be done in the space provided for this purpose in the Test Booklet only.
5. On completion of the test, the candidate **must hand over the Answer Sheet (ORIGINAL and OFFICE Copy)** to the **Invigilator** before leaving the room/hall. The candidates are allowed to take away this Test Booklet with them.
6. The CODE for this Booklet is **12**.
7. The candidates should ensure that the Answer Sheet is not folded. Do not make any stray marks on the Answer Sheet. Do not write your Roll No. anywhere else except in the specified space in the Test Booklet/Answer Sheet. The use of white fluid for correction is **NOT** permissible on the Answer Sheet.
8. Each candidate must show on-demand his/her Admit Card to the Invigilator.
9. No candidate, without special permission of the Centre Superintendent or Invigilator, would leave his/her seat.
10. Use of an Electronic/Manual Calculator is prohibited.
11. The candidates are governed by all Rules and Regulations of the examination with regard to their conduct in the Examination Room/Hall. All cases of unfair means will be dealt with as per Rules and Regulations of this examination.
12. No part of the Test Booklet and Answer Sheet shall be detached under any circumstances.
13. The candidates will write the Correct Test Booklet Code as given in the Test Booklet/Answer Sheet in the Attendance Sheet.

1. A 100-turn closely wound circular coil of radius 5 cm has a magnetic field of 3.14×10^{-3} T at its centre. The current flowing through the coil, and the magnitude of the magnetic moment of this coil are, respectively: (Take $\mu_0 = 4\pi \times 10^{-7}$ T m/A)

- (1) 2.5 A, 2 A m²
- (2) 2.5 A, 20 A m²
- (3) 2 A, 4 A m²
- (4) 2 A, 10 A m²

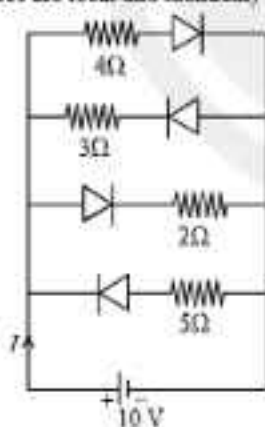
2. Match List-I with List-II.

List-I	List-II
(A) $E = h\nu$	(I) de Broglie wavelength
(B) Diffraction and Interference	(II) Particle nature of light
(C) $\lambda = \frac{h}{p}$	(III) Wave nature of light
(D) Compton effect	(IV) Energy of photon

Choose the correct answer from the options given below:

- (1) A-IV, B-III, C-I, D-II
- (2) A-I, B-IV, C-III, D-II
- (3) A-IV, B-I, C-II, D-III
- (4) A-IV, B-III, C-II, D-I

3. The current I in the circuit shown below is: (All diodes are ideal and identical)

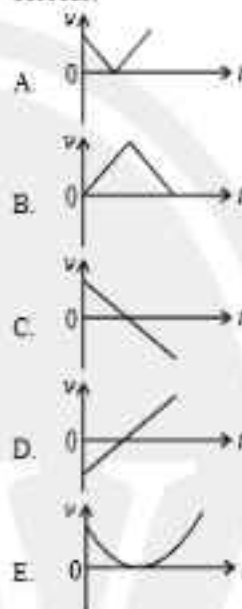


- (1) $\frac{5}{3}$ A
- (2) $\frac{15}{2}$ A
- (3) $\frac{1}{3}$ A
- (4) $\frac{5}{9}$ A

4. The speed of light in vacuum is taken as unity. If light takes 6 min 40 s to reach the Earth from the Sun, the distance between the Sun and the Earth in new unit is:

- (1) 3×10^8
- (2) 3×10^{10}
- (3) 400
- (4) 500

5. The following plots show variation of velocity (v) with time (t), of a ball thrown vertically upward, and falling back. Which of the following plots is/are correct?



- (1) C only
- (2) A and E only
- (3) D only
- (4) B only

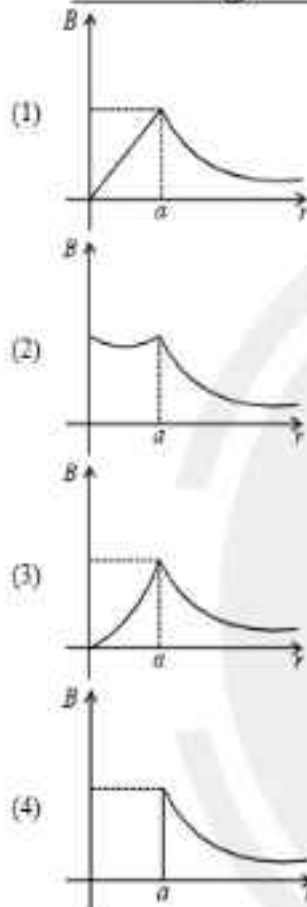
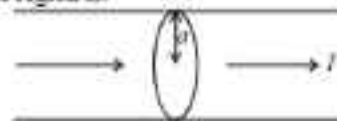
6. In a vernier callipers, 20 VSD coincide with 16 MSD (each division of length 1 mm). The least count of the vernier callipers is:

- (1) 0.01 cm
- (2) 0.1 cm
- (3) 0.02 cm
- (4) 0.2 cm

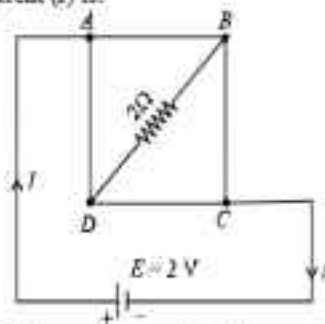
7. An ac circuit contains a resistance of $1\text{ k}\Omega$, a capacitor of $0.1\text{ }\mu\text{F}$ and an inductor of 1 mH connected in series. The resonance frequency of the circuit is approximately:

- (1) 10.1 kHz
- (2) 20.7 kHz
- (3) 15.9 kHz
- (4) 13.5 kHz

8. The figure given below shows a long straight solid wire of circular cross-section of radius ' a ' carrying steady current I . The current I is uniformly distributed across its cross-section. The plot which correctly represents the variation of magnetic field (B) with distance (r) from the axis of the conductor in the region is:



9. A uniform metallic wire having resistance $4\ \Omega$ is bent to form a square loop ($ABCD$) (see figure). A resistance of $2\ \Omega$ is connected between points B and D and a battery of 2 V is connected across points A and C as shown in the figure. Now the value of current (I) is:



- (1) 2 A (2) 4 A
(3) 8 A (4) 4.5 A

10. An unknown nucleus has a nuclear density of $2.29 \times 10^{17}\text{ kg/m}^3$ and mass of $19.926 \times 10^{-27}\text{ kg}$. Its mass number A is approximately :

(Take $R_0 = 1.2 \times 10^{-15}\text{ m}$, $4\pi = 12.56$)

- (1) 12
(2) 19
(3) 20
(4) 16

11. A rectangular wire loop of sides 8 cm and 3 cm with a small cut, is moving out of a region of uniform magnetic field of magnitude 0.3 T directed normal to the plane of the loop. The emf developed across the cut, if the velocity of the loop is 2 cm s^{-1} , in a direction normal to the shorter side of the loop, will be :

- (1) $1.8 \times 10^{-4}\text{ volt}$
(2) $1.2 \times 10^{-4}\text{ volt}$
(3) $1.3 \times 10^{-4}\text{ volt}$
(4) $4.8 \times 10^{-4}\text{ volt}$

12. A galvanometer of resistance $100\ \Omega$ gives full scale deflection for a current of 1 mA . It is converted into an ammeter of range $0 - 10\text{ A}$. The shunt required is:

- (1) $0.01\ \Omega$
(2) $0.10\ \Omega$
(3) $0.001\ \Omega$
(4) $1.0\ \Omega$

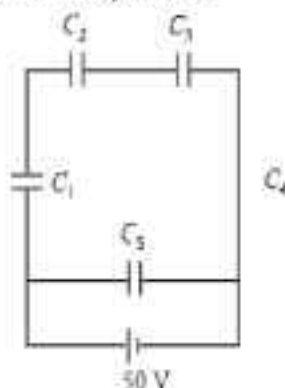
13. In Young's double slit experiment, using monochromatic light of wavelength λ , the intensity of light at a point on the screen where the path difference is λ is K units. The intensity of light at a point where the path difference is $\frac{\lambda}{3}$ will be :

- (1) $\frac{K}{4}$ (2) K
(3) $\frac{K}{2}$ (4) $2K$

14. The magnitude and direction of the acceleration produced in a body of mass 5 kg when two mutually perpendicular forces 8 N and 6 N act on it, are respectively :

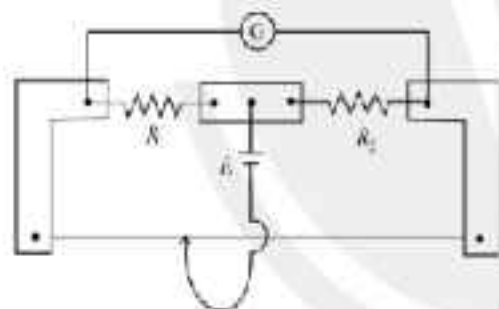
- (1) 2 m s^{-2} ; $\tan^{-1}(3/4)$ with 6 N force
(2) 2 m s^{-2} ; $\tan^{-1}(4/3)$ with 8 N force
(3) 2 m s^{-2} ; $\tan^{-1}(3/4)$ with 8 N force
(4) 20 m s^{-2} ; $\tan^{-1}(4/3)$ with 8 N force

15. Five capacitors of capacitances $C_1 = C_2 = C_3 = C_4 = 10 \mu\text{F}$ and $C_5 = 2.5 \mu\text{F}$ are connected as shown, along with a battery of 50 V.



The equivalent capacitance and the charges on each capacitor respectively are:

- (1) $5 \mu\text{F}$, $125 \mu\text{C}$ on all capacitors
 - (2) $5 \mu\text{F}$, $250 \mu\text{C}$ on all capacitors
 - (3) $4 \mu\text{F}$, $250 \mu\text{C}$ on C_1 to C_4 and $125 \mu\text{C}$ on C_5
 - (4) $5 \mu\text{F}$, $125 \mu\text{C}$ on C_1 to C_4 and $25 \mu\text{C}$ on C_5
16. In a metre bridge experiment (see figure), the positions of the cell, E, and galvanometer, G, are interchanged. We shall observe in the galvanometer:



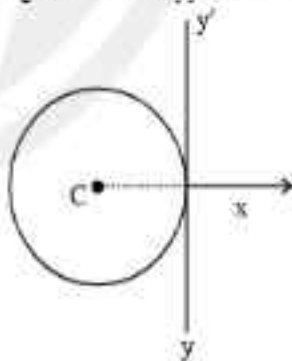
- (1) Only the right-sided deflection
 - (2) Only the left-sided deflection
 - (3) There will be no deflection irrespective of the position of the jockey
 - (4) Both right-sided and left-sided deflection and at balance point, no-deflection
17. The power of a crane, which lifts a mass of 1000 kg to a height of 20 m in 10 s is : ($g = 9.8 \text{ m/s}^2$)
- (1) 19.6 W
 - (2) 39.2 W
 - (3) 39.2 kW
 - (4) 19.6 kW

18. Match List-I with List-II

List-I		List-II	
(A)	Young's Modulus	(I)	$\frac{\Delta d}{\Delta L} \left(\frac{L}{d} \right)$
(B)	Compressibility	(II)	$\frac{FL}{A(\Delta L)}$
(C)	Bulk Modulus	(III)	$-\frac{1}{\Delta P} \left(\frac{\Delta V}{V} \right)$
(D)	Poisson's Ratio	(IV)	$-P \left(\frac{V}{\Delta V} \right)$

Choose the correct answer from the options given below:

- (1) A-I, B-IV, C-III, D-II
 - (2) A-IV, B-I, C-II, D-III
 - (3) A-III, B-II, C-I, D-IV
 - (4) A-II, B-III, C-IV, D-I
19. In a concave lens, a ray of light emanating from the object parallel to the principal axis of the lens, after refraction:
- (1) emerges parallel to the principal axis.
 - (2) appears to diverge from the first principal focus.
 - (3) passes through $2F$, which is the radius of curvature of the lens.
 - (4) passes through the second principal focus.
20. A thin wire of length 'L' and linear mass density 'm' is bent into a circular ring (in $x-y$ plane) with centre 'C' as shown in figure. The moment of inertia of the ring about an axis yy' will be:



- (1) $\frac{3mL^3}{8\pi^2}$
- (2) $\frac{3mL^3}{8\pi}$
- (3) $\frac{3mL^2}{8\pi^2}$
- (4) $\frac{3mL^2}{8\pi}$

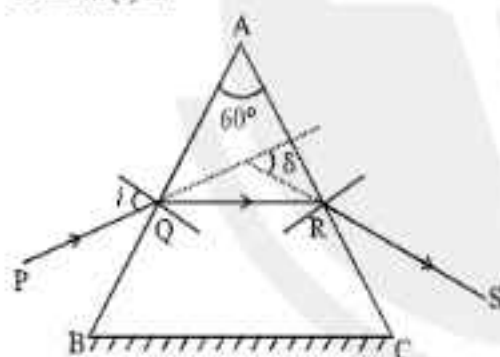
21. Each side of a metallic cube of mass 5.580 kg is measured to be 9.0 cm. Keeping the significant figures in view, the density of the material of the cube can be best expressed as $X \times 10^3 \text{ kg m}^{-3}$, where the value of X is:

(1) 7.654
(2) 7.7
(3) 7.65
(4) 7.6

22. For a travelling harmonic wave $y(x, t) = 2.0 \cos 2\pi(10t - 0.0080x + 0.35)$, where x and y are in cm and t in s. The phase difference between oscillatory motion of two points separated by a distance of 0.5 m is:

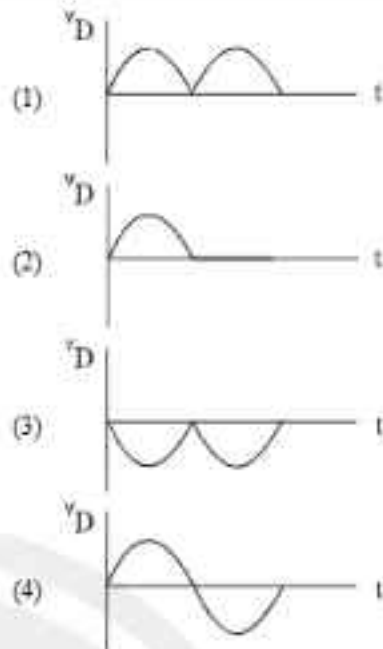
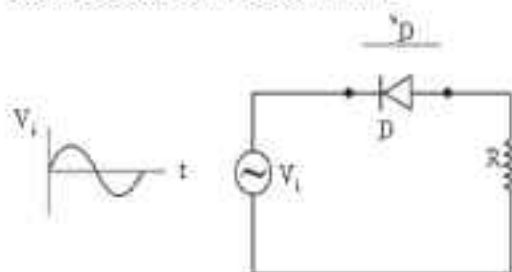
(1) 8π rad
(2) 0.08π rad
(3) 0.008π rad
(4) 0.8π rad

23. A ray of monochromatic light is passing through an equilateral prism (ABC) as shown in the figure. The refracted ray (QR) is parallel to its base (BC) and the angle of incidence (i) is 50° . Then the angle of deviation (δ) is:

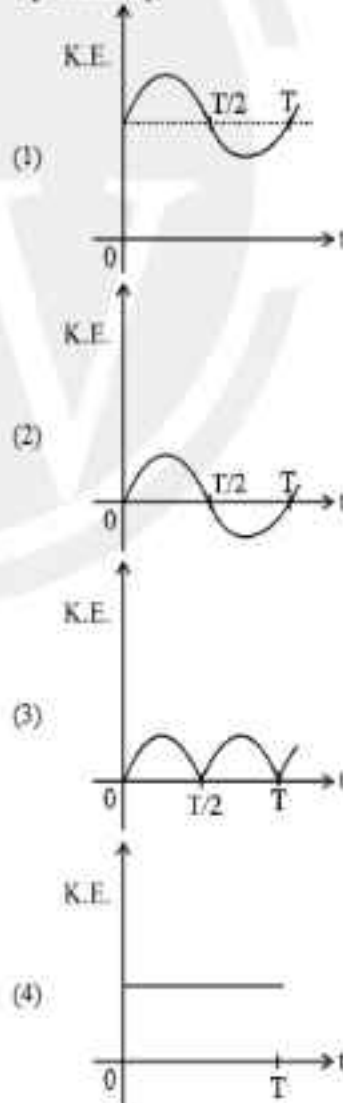


(1) 40°
(2) 45°
(3) 55°
(4) 35°

24. In the circuit shown below, the voltage appearing across the diode D will be of the form:



25. For a simple pendulum, having time period T , the variation of kinetic energy (K.E.) with time (t) is represented by:



26. A resistor is connected to a battery of 12 V emf and internal resistance $2\ \Omega$. If the current in the circuit is 0.6 A, the terminal voltage of the battery is:
 (1) 10 V (2) 10.8 V
 (3) 12 V (4) 1.2 V
27. The amount of work done to raise a mass 'm' from the surface of the Earth to a height equal to the radius of the Earth 'R', will be:
 (1) $2\ mg\ R$ (2) $mg\ R$
 (3) $mg\ \frac{R}{4}$ (4) $mg\ \frac{R}{2}$
28. An electric heater supplies heat to a system at a rate of 100 W. If the system performs work at a rate of 75 J/s, then the rate at which internal energy increases will be:
 (1) 125 W (2) 100 W
 (3) 25 W (4) 75 W
29. A room heater is rated 400 W, 220 V. If the supply voltage drops to 200 V, what will be the power consumed (approximately)?
 (1) 121 W (2) 331 W
 (3) 200 W (4) 400 W
30. When a ruler falls vertically, 5 different persons catch it with different reaction times.
 ($g = 9.8\ \text{ms}^{-2}$)
 A. Person A has reaction time of 0.20 s.
 B. Person B has reaction time of 0.22 s.
 C. Person C has reaction time of 0.18 s.
 D. Person D has reaction time of 0.19 s.
 E. Person E has reaction time of 0.21 s.
 What is the correct order of the distance travelled by the ruler for each person?
 (1) $C > D > A > B > E$
 (2) $C > D > A > E > B$
 (3) $B > E > A > C > D$
 (4) $B > E > A > D > C$
31. Consider two uncharged capacitors of equal capacitance 200 pF. One of them is charged by a 100 V supply and disconnected. Now this capacitor is connected to the uncharged capacitor. The amount of electrostatic energy lost in the process is:
 (1) $1.0 \times 10^{-4}\ \text{J}$ (2) $0.5 \times 10^{-4}\ \text{J}$
 (3) $0.5\ \text{J}$ (4) $1.0\ \text{J}$
32. Savitha, a XI standard student, while conducting an experiment to determine the effective length of a simple pendulum L, notes down the data of time taken to complete 30 oscillations as 60 s and hence calculates the length of the simple pendulum as:
 (Take $\pi^2 = 9.8$, and $g = 9.8\ \text{m/s}^2$)
 (1) 2 m (2) 0.75 m
 (3) 1.5 m (4) 1 m
33. The peak value of an alternating current is 5 A and frequency is 60 Hz. How long will the current, starting from zero, take to reach the peak value?
 (1) $\frac{1}{240}\ \text{s}$ (2) $\frac{1}{30}\ \text{s}$
 (3) $\frac{1}{120}\ \text{s}$ (4) $\frac{1}{60}\ \text{s}$
34. In interference and diffraction, the light energy is redistributed. If it reduces in one region, producing a dark fringe, it increases in another region, producing a bright fringe.
 A. As there is no gain or loss of energy, these phenomena are consistent with the principle of conservation of energy.
 B. Diffraction and interference are characteristics exhibited only by light waves.
 Choose the correct answer from the options given below:
 (1) A is true, but B is false
 (2) A is true and B is also true
 (3) A is false, but B is true
 (4) Both A and B are false
35. A box of mass 15 kg is kept on the floor of a stationary trolley. The coefficient of static friction between the box and the trolley is 0.12. Keeping the box in stationary state over the trolley, the maximum acceleration with which the trolley can be moved horizontally in m s^{-2} is:
 ($g = 10\ \text{m/s}^2$)
 (1) 1.5 (2) 1.8
 (3) 2.1 (4) 1.2
36. The sum of kinetic energy and potential energy of a simple pendulum bob is 0.02 joule. The speed of the simple pendulum bob at equilibrium position is approximately:
 (Consider mass of the bob = 20 g)
 (1) 1.41 m/s (2) 14.1 m/s
 (3) 0.2 m/s (4) 2.0 m/s
37. Four statements are given (A is mass number):
 A. The volume of a nucleus is proportional to $A^{3/3}$.
 B. The volume of a nucleus is proportional to A.
 C. The difference in mass of an atom and its nucleus is called the mass defect.
 D. The difference in mass of a nucleus and its constituents is called the mass defect.
 Choose the correct answer from the options given below:
 (1) B and D are true, but A and C are false
 (2) A and D are true, but B and C are false
 (3) A and C are true, but B and D are false
 (4) B and C are true, but A and D are false

38. The angular speed of a flywheel is increased from 600 rpm to 1200 rpm in 10 s. The number of revolutions completed by the flywheel during this time is :

(1) 600 (2) 900
(3) 300 (4) 150

39. A submarine is designed to withstand an absolute pressure of 100 atm. How deep can it go below the water surface ?

(Consider the density of water = 1000 kg m^{-3} , $1 \text{ atm} = 1 \times 10^5 \text{ Pa}$ and gravitational acceleration $g = 10 \text{ m/s}^2$)

(1) 9900 m (2) 99 m
(3) 9000 m (4) 990 m

40. Match List I with List II :

List I (Electromagnetic wave)		List II (Production)	
A.	Microwave	I.	Electrons in atoms emit light when they move from a higher energy level to a lower energy level
B.	Visible light	II.	Radioactive decay of nucleus
C.	Gamma rays	III.	Vibration of atoms and molecules
D.	Infra-red rays	IV.	Klystron valve or magnetron valve

Choose the correct answer from the options given below :

(1) A-III, B-I, C-II, D-IV
(2) A-III, B-IV, C-I, D-II
(3) A-IV, B-III, C-II, D-I
(4) A-IV, B-I, C-II, D-III

41. Which of the following statements are correct ?

A. Inside a conductor, the electrostatic field is zero.
B. Electric field at the surface of a charged conductor does not depend on its surface charge density.
C. The interior of a charged conductor can have no excess charge in the static situation.
D. At the surface of a charged conductor, the electrostatic field must be normal to the surface at every point.
E. The electrostatic potential is zero everywhere inside a charged conductor.

Choose the correct answer from the options given below :

(1) C, D and E only
(2) A, B and D only
(3) A, C and D only
(4) A, C and E only

42. For a metal of work function 6.6 eV, which of the following wavelengths of incident radiation does not give rise to the photoelectric effect ?

(Take Planck's constant as $6.6 \times 10^{-34} \text{ J s}$)

(1) 200 nm (2) 150 nm
(3) 100 nm (4) 50 nm

43. In the first excited state of hydrogen atom, the energy of its electron is -3.4 eV. The radial distance of the electron from the hydrogen nucleus in this case is approximately :

(Take $1 \text{ eV} = 1.6 \times 10^{-19} \text{ J}$, $e = 1.6 \times 10^{-19} \text{ C}$ and $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9 \text{ N m}^2/\text{C}^2$)

(1) $2.1 \times 10^{-8} \text{ m}$ (2) $2.1 \times 10^{-11} \text{ m}$
(3) $2.1 \times 10^{-9} \text{ m}$ (4) $2.1 \times 10^{-10} \text{ m}$

44. Two statements are given below :

A. When the forward bias voltage across a p-n junction diode increases above a certain threshold voltage, the diode current increases significantly.

B. This current is called reverse saturation current.

Choose the correct answer from the options given below :

(1) Both Statements A and B are false
(2) Statement A is true, but Statement B is false
(3) Both Statements A and B are true
(4) Statement A is false, but Statement B is true

45. A flask contains argon and chlorine in the ratio of 2 : 1 by mass. The temperature of the mixture is 27°C . The ratio of root mean square speed of the

molecules of the two gases $\left(\frac{v_{\text{rms}}^{\text{Ar}}}{v_{\text{rms}}^{\text{Cl}}}\right)$ is :

(Atomic mass of argon = 40.0 u and molecular mass of chlorine = 70.0 u)

(1) $\frac{\sqrt{7}}{2}$ (2) $\frac{7}{2}$
(3) $\frac{7}{4}$ (4) $\frac{2}{\sqrt{7}}$

53. Match List-I with List-II

List-I (Transition metal/ compound/complex)		List-II (Catalytic Role)	
(A)	V_2O_5	(I)	Preparation of ammonia from N_2/H_2 mixture
(B)	Fe	(II)	Polymerisation of alkynes
(C)	$PdCl_2$	(III)	Preparation of H_2SO_4 from SO_2
(D)	Ni complex	(IV)	Oxidation of ethyne to ethanal

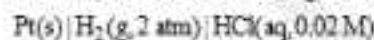
Choose the correct answer from the options given below:

- (1) A-III, B-IV, C-I, D-II
 (2) A-II, B-I, C-IV, D-III
 (3) A-IV, B-I, C-III, D-II
 (4) A-III, B-I, C-IV, D-II

54. Identify the correct statement about ClF_3 from the following options:

- (1) It has a trigonal pyramidal geometry with two lone pairs on Cl atom.
 (2) It has T-shaped geometry with two lone pairs on Cl atom.
 (3) It has a planar trigonal geometry with two lone pairs on Cl atom.
 (4) It has T-shaped geometry with three lone pairs on Cl atom.

55. Calculate emf of the half cell given below:



$$E^\circ_{H_2/H^+} = 0 \text{ V}$$

$$\left(\text{Given : } \frac{2.303 RT}{F} = 0.059, \log 2 = 0.3010 \right)$$

- (1) 0.109 V (2) 0.035 V
 (3) -0.035 V (4) -0.109 V

56. Match the List-I with List-II

List-I (Order of reaction)		List-II (Unit of rate constant)	
(A)	Zero order	(I)	$\text{mol}^{-1} \text{L s}^{-1}$
(B)	First order	(II)	$\text{mol}^{-2} \text{L}^2 \text{s}^{-1}$
(C)	Second order	(III)	s^{-1}
(D)	Third order	(IV)	$\text{mol L}^{-1} \text{s}^{-1}$

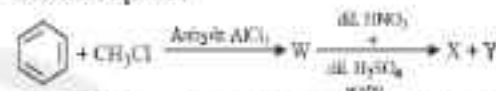
Choose the correct answer from the options given below:

- (1) A-IV, B-III, C-II, D-I
 (2) A-I, B-II, C-III, D-IV
 (3) A-IV, B-III, C-I, D-II
 (4) A-IV, B-II, C-I, D-III

57. The calculated 'spin-only' magnetic moment of $Ti^{2+}(3d^2)$ is:

- (1) 2.84 BM (2) 5.92 BM
 (3) 4.90 BM (4) 3.87 BM

58. Two products X and Y are formed in the following reaction sequence.



The suitable method that can be used for the separation of products X and Y is:

- (1) Continuous extraction
 (2) Differential extraction
 (3) Fractional distillation
 (4) Sublimation

59. A bulb is rated at 150 watt, converting 8% energy into light. If energy of one photon is $4.42 \times 10^{-19} \text{ J}$, how many photons are emitted by the bulb per second?

- (1) 1.35×10^{19} (2) 4.06×10^{19}
 (3) 2.71×10^{19} (4) 27.2×10^{19}

60. In a test tube containing a salt, a few drops of dilute H_2SO_4 was added, which gave colourless vapours having the smell of vinegar. The vapours turned the blue litmus paper red.

Identify the correct anion from the following:

- (1) Acetate, CH_3COO^-
 (2) Carbonate, CO_3^{2-}
 (3) Sulphate, SO_4^{2-}
 (4) Sulphide, S^{2-}

61. Select the reagents that reduce nitriles to primary amines:

- A. (i) LiAlH_4 ; (ii) H_2O
 B. $\text{Sn} + \text{HCl}$
 C. H_2/Ni
 D. $\text{Na}(\text{Hg})/\text{C}_2\text{H}_5\text{OH}$
 E. $\text{BCl}_3/\text{aq. NaOH}$

Choose the correct answer from the options given below:

- (1) A, B and C only
 (2) A, C and D only
 (3) A, D and E only
 (4) B, D and E only

62. Identify the **incorrect** statement from the following:
- (1) Carbon has the ability to form $pn-pn$ multiple bond with itself.
 - (2) ECl_3 ($E = B$ and Al) is a monomer when $E = B$ and a dimer when $E = Al$.
 - (3) Oxygen exhibits only -2 oxidation state.
 - (4) The order of catenation property of Group 14 elements is $C \gg Si > Ge \approx Sn$.
63. Although $+3$ oxidation state is most common in lanthanoids, cerium still shows $+4$ oxidation state because:
- (1) Its nearest gas is Radon.
 - (2) After losing one more electron, it acquires $4f^4$ electronic configuration.
 - (3) Its atomic number is 61.
 - (4) After losing one more electron, it acquires $4f^0$ electronic configuration.
64. During Lassaigne's test, the elements present in an organic compound are converted from :
- (1) covalent form to covalent form
 - (2) ionic form to ionic form
 - (3) covalent form to ionic form
 - (4) ionic form to covalent form
65. The number of hydrogen atoms present in 5.4 g of urea is :
- (Given : Molar mass of urea : 60 g mol^{-1} , $N_A : 6.022 \times 10^{23} \text{ particles mol}^{-1}$)
- (1) 2.168×10^{22}
 - (2) 2.168×10^{23}
 - (3) 1.084×10^{22}
 - (4) 1.084×10^{23}
66. The pair of molecules that are metamers among the following is :
- (1) $CH_3CH_2CH_2OH$ and $CH_3-CH(OH)-CH_3$
 - (2) $CH_3OCH_2CH_2CH_3$ and $CH_3CH_2OCH_2CH_3$
 - (3) $H_3C-C(=O)-CH_3$ and $H_3C-CH_2-C(=O)-H$
 - (4) $CH_3CH_2CH_2CH_2CH_3$ and $(CH_3)_2CHCH_2CH_3$
67. Identify the **incorrect** statement from the following :
- (1) $P(C_2H_5)_3$ and $As(C_2H_5)_3$ form $dx-dx$ bond with transition metals.
 - (2) Nitrogen can form $dx-p\pi$ bond with oxygen.
 - (3) Nitrogen can form $pn-pn$ multiple bonds with itself.
 - (4) Phosphorus, arsenic and antimony show catenation property.

68. Phenolphthalein is used as an indicator for the titration of sodium hydroxide solution against a standard solution of oxalic acid. The colour change that is observed at an alkaline pH close to the equivalence point during this titration is :

- (1) pinkish red to yellow
- (2) yellow to pinkish red
- (3) colourless to pink
- (4) pink to colourless

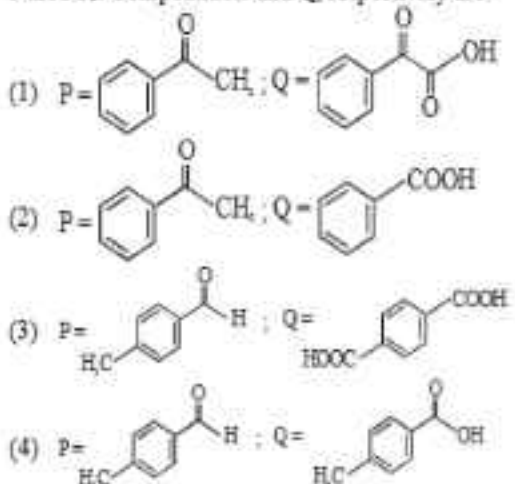
69. Match List I with List II :

List-I		List-II	
A.	C_2H_4	I.	3 σ bonds, 2 π bonds
B.	C_2H_2	II.	3 σ bonds, one lone pair
C.	CH_4	III.	4 σ bonds
D.	NH_3	IV.	5 σ bonds, 1 π bond

Choose the **correct** answer from the options given below :

- (1) A-IV, B-I, C-III, D-II
 - (2) A-III, B-IV, C-II, D-I
 - (3) A-I, B-II, C-IV, D-III
 - (4) A-II, B-III, C-I, D-IV
70. At a certain temperature, T (K), during a process, 500 J is absorbed by the system and work of 200 J is done by the system. Then change in internal energy of the system is :
- (1) 700 J
 - (2) 300 J
 - (3) 400 J
 - (4) 500 J
71. Methane reacts with steam at 1273 K in the presence of nickel catalyst to form :
- (1) CO and H_2
 - (2) CO and H_2O
 - (3) CO_2 and H_2O
 - (4) CO_2 and H_2

72. Compound P (C_8H_8O) gives a red orange precipitate with 2,4-DNP reagent and it does not reduce Fehling's reagent. On drastic oxidation with chromic acid, P gives an aromatic product Q that produces effervescence on treating with aq. $NaHCO_3$. Compounds P and Q, respectively are:



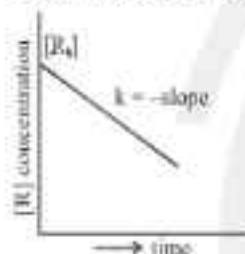
73. A solution of copper sulphate is electrolysed for 10 minutes with a current of 1.5 amperes. The mass of copper deposited at cathode is :
(Given : Molar mass of Cu = 63 g mol^{-1} ; $1F = 96487 \text{ C mol}^{-1}$)
- (1) 2.4036 g
 - (2) 1.7018 g
 - (3) 0.5876 g
 - (4) 0.2938 g
74. The functional group that can be identified through phthalein dye test is :
- (1) Phenolic
 - (2) Alcohol
 - (3) Aldehyde
 - (4) Carboxylic acid
75. The correct statement with regard to the secondary structure of DNA/RNA is :
- (1) DNA possesses a single strand helix structure and contains uracil as one of the four bases.
 - (2) RNA possesses a single strand helix structure and contains thymine as one of the four bases.
 - (3) DNA possesses a double strand helix structure and contains thymine as one of the four bases.
 - (4) RNA possesses a double strand helix structure and contains uracil as one of the four bases.
76. Identify the correct statements :
- A. The molality of 2.5 g of ethanoic acid (Molar mass : 60 g mol^{-1}) in 75 g of benzene solution is 0.556 m.
 - B. The molarity of a solution containing 5 g of NaOH (molar mass : 40 g mol^{-1}) in 450 mL of solution is 0.278 M at 298 K.
 - C. Aquatic species are more comfortable in cold water.
 - D. The solubility of gas increases with decrease in pressure.
 - E. For a binary mixture of A and B, the number of moles of A and B are n_A and n_B respectively.
The mole fraction of B will be $x_B = \frac{n_B}{n_A + n_B}$.
- Choose the correct answer from the options given below :
- (1) A and C only
 - (2) A, B and C only
 - (3) A, D and E only
 - (4) A and B only
77. Mixture of chloroform and acetone forms a solution with negative deviation from Raoult's law due to :
- (1) formation of hydrogen bonding between acetone and chloroform.
 - (2) increase in escaping tendency of molecules of each component.
 - (3) stronger intermolecular forces between chloroform molecules than those between chloroform and acetone molecules.
 - (4) repulsive forces.
78. At 298 K, a certain buffer solution contains equal concentrations of X^- and HX . K_a for X^- is 10^{-10} . What is the pH of this buffer solution ?
- (1) 2
 - (2) 10
 - (3) 4
 - (4) 6
79. Identify the incorrect statement from the following:
- (1) The IUPAC name of the element with atomic number 107 is Unnilseptium.
 - (2) The largest and the smallest species among Mg , Mg^{2+} , Al and Al^{3+} are Al and Mg^{2+} , respectively.
 - (3) The similarity in behaviour of Li with Mg is referred to as 'diagonal relationship'.
 - (4) The oxidation state and covalency of Al in $[AlCl(H_2O)_5]^{2+}$ are 3 and 6 respectively.
80. The correct order of increasing metallic character of Na, Be, P, Mg and Si is :
- (1) $P < Si < Be < Mg < Na$
 - (2) $Be < Si < P < Mg < Na$
 - (3) $P < Si < Na < Mg < Be$
 - (4) $P < Mg < Be < Si < Na$
81. The correct IUPAC name of the following compound is :
- $$\begin{array}{ccccccc}
 \text{CH}_3 & - & \text{CH}_2 & - & \text{CH} & - & \text{CH}_2 & - & \text{CH} & - & \text{CH}_2 & - & \text{CH}_3 \\
 & & & & | & & & & | & & & & \\
 & & & & \text{CH}_3 & & & & \text{CH}_3 & & & &
 \end{array}$$
- (1) 2,4-diethylhexane
 - (2) 3,5-diethylhexane
 - (3) 3-ethyl-5-methylheptane
 - (4) 3-methyl-5-ethylheptane

82. Match List I with List II:

List-I (Complex/ion)		List-II (Shape/geometry)	
A.	$[\text{Pt}(\text{Cl}_2)(\text{NH}_3)_2]$	I.	Octahedral
B.	$[\text{Co}(\text{NH}_3)_5\text{Cl}]$	II.	Trigonal bipyramidal
C.	$[\text{NiCl}_4]^{2-}$	III.	Square planar
D.	$[\text{Fe}(\text{CO})_5]$	IV.	Tetrahedral

Choose the correct answer from the options given below:

- (1) A-I, B-III, C-IV, D-II
 (2) A-III, B-IV, C-I, D-II
 (3) A-IV, B-I, C-III, D-II
 (4) A-III, B-I, C-IV, D-II

83. For a certain reaction $R \rightarrow \text{Product}$, the plot of concentration $[R]$ vs time has a negative slope as shown. The order of reaction is:

- (1) 0
 (2) 1
 (3) 2
 (4) 2.5

84. Which one of the following is an ambidentate ligand?

- (1) Ethylenediaminetetraacetate ion
 (2) Oxalate
 (3) Ethane-1,2-diamine
 (4) Thiocyanate

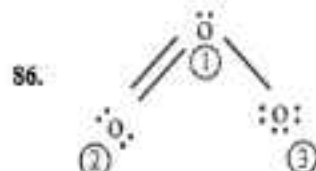
85. Consider the following reaction:



$$\Delta U^\circ = -10 \text{ kJ mol}^{-1} \text{ and } \Delta S^\circ = -44 \text{ J K}^{-1} \text{ at } 298 \text{ K}$$

Identify the correct option with ΔG° for the reaction and spontaneity of the reaction at 298 K.(Given: $R = 8.31 \text{ J mol}^{-1} \text{ K}^{-1}$)

- (1) $-1.635 \text{ kJ mol}^{-1}$, spontaneous
 (2) $+0.63568 \text{ kJ mol}^{-1}$, non-spontaneous
 (3) $-0.63568 \text{ kJ mol}^{-1}$, spontaneous
 (4) $+1.635 \text{ kJ mol}^{-1}$ non-spontaneous



86.

The correct formal charges on oxygen atoms numbered 2, 1 and 3 respectively are:

- (1) $-1, 0, +1$ (2) $0, -1, -1$
 (3) $0, 0, 0$ (4) $+1, 0, -1$

87. Given below are certain reactions. Identify the reaction for which $K_p \neq K_c$.

- (1) $\text{H}_2(\text{g}) + \text{I}_2(\text{g}) \rightleftharpoons 2\text{HI}(\text{g})$
 (2) $\text{N}_2(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons 2\text{NO}(\text{g})$
 (3) $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightleftharpoons 2\text{NH}_3(\text{g})$
 (4) $\text{H}_2\text{O}(\text{g}) + \text{CO}(\text{g}) \rightleftharpoons \text{H}_2(\text{g}) + \text{CO}_2(\text{g})$

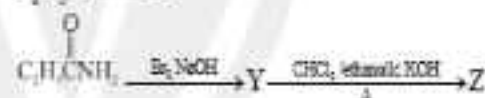
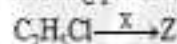
88. Given below is an expression for the rate constant of a first order reaction occurring at a certain temperature, T (K).

$$\ln k = 14.34 - \frac{1.25 \times 10^4}{T}$$

The energy of activation in kcal mol^{-1} for the reaction is:(Given: k in s^{-1} , $R = 1.987 \text{ cal mol}^{-1} \text{ K}^{-1}$)

- (1) 12.42 (2) 14.34
 (3) 18.63 (4) 24.84

89. The following two reactions give the same foul smelling product Z.



X and Z, respectively, are:

- (1) $\text{X} = \text{AgCN}$; $\text{Z} = \text{C}_2\text{H}_5\text{CN}$
 (2) $\text{X} = \text{KCN}$; $\text{Z} = \text{C}_2\text{H}_5\text{CN}$
 (3) $\text{X} = \text{KCN}$; $\text{Z} = \text{C}_2\text{H}_5\text{NC}$
 (4) $\text{X} = \text{AgCN}$; $\text{Z} = \text{C}_2\text{H}_5\text{NC}$

90. Match List I with List II:

List I (Complex)	List II (Type of isomerism)
A. $[\text{Pt}(\text{NH}_3)_2\text{Cl}_2]$	I. Optical
B. $[\text{Co}(\text{en})_3]^{3+}$	II. Solvate
C. $[\text{Co}(\text{NH}_3)_5\text{NO}_2]\text{Cl}_2$	III. Geometrical
D. $[\text{Cr}(\text{H}_2\text{O})_6]\text{Cl}_2$	IV. Linkage

Choose the correct answer from the options given below:

- (1) A-III, B-I, C-II, D-IV
 (2) A-I, B-III, C-II, D-IV
 (3) A-III, B-I, C-IV, D-II
 (4) A-II, B-IV, C-III, D-I

91. "The Evil Quartet" of biodiversity loss includes which of the following?

- (1) Over-exploitation; Alien species invasions; Air pollution; Co-extinctions
- (2) Habitat loss and fragmentation; over-exploitation; Alien species invasions; Co-extinctions
- (3) Habitat loss and fragmentation; Air pollution; Water pollution; Co-extinctions
- (4) Over-exploitation; Alien species invasions; Soil pollution; Co-extinctions

92. Which one of the following is the site for active ribosomal RNA synthesis?

- (1) Nucleolus
- (2) Chromatin
- (3) Centrosome
- (4) Kinetochore

93. Match List-I with List-II:

List-I (Phase of cell cycle)		List-II (Activity)	
(A)	G ₁ phase	(I)	Actual cell division occurs
(B)	S phase	(II)	Cell is metabolically active and continuously grows but does not replicate its DNA
(C)	G ₂ phase	(III)	Synthesis of DNA occurs and the amount of DNA per cell doubles
(D)	M phase	(IV)	Proteins are synthesized while cell growth continues

Choose the correct answer from the options given below:

- (1) A-II, B-III, C-IV, D-I
- (2) A-III, B-IV, C-I, D-II
- (3) A-I, B-II, C-III, D-IV
- (4) A-IV, B-I, C-II, D-III

94. Match List-I with List-II:

List-I		List-II	
(A)	Productivity	(I)	Gross primary productivity minus respiration losses
(B)	Net primary productivity	(II)	Rate of formation of new organic matter by consumers
(C)	Gross primary productivity	(III)	Rate of biomass production
(D)	Secondary productivity	(IV)	Rate of production of organic matter during photosynthesis

Choose the correct answer from the options given below:

- (1) A-I, B-II, C-III, D-IV
- (2) A-III, B-I, C-IV, D-II
- (3) A-III, B-I, C-II, D-IV
- (4) A-I, B-III, C-IV, D-II

95. Which of the following statements are correct?

- A. The Amazon rainforest being cut and cleared for cultivation of soyabean is an example of habitat loss.
- B. Steller's sea cow and passenger pigeon became extinct due to over-exploitation by humans.
- C. The Nile perch introduced into Lake Victoria in East Africa helped in population growth of cichlid fish in the lake.
- D. Water hyacinth is an invasive species.
- E. When a species becomes extinct, the plant and animal species associated with it are not affected.

Choose the correct answer from the options given below:

- (1) B, C and D only
- (2) A, B and D only
- (3) A, B and E only
- (4) C, D and E only

96. Identify the correct statements about biomolecules.
- Lipids are generally water soluble.
 - Proteins are polypeptides.
 - Polysaccharides are long chains of sugars.
 - Adenine and guanine are substituted pyrimidines.
 - Almost all enzymes are proteins.

Choose the **correct** answer from the options given below:

- (1) C, D and E only (2) B, C and E only
(3) B, D and E only (4) A, B and C only

97. How many ATP and NADPH molecules are required to make one molecule of glucose through the Calvin pathway?

- (1) 18 ATP and 12 NADPH
(2) 6 ATP and 12 NADPH
(3) 24 ATP and 18 NADPH
(4) 12 ATP and 18 NADPH

98. Which of the following statements are **not** true regarding restriction endonucleases?

- They are called molecular scissors.
- These are the enzymes responsible for restricting the growth of bacteriophages in *E. coli*.
- They cut the DNA only at the centre of the palindromic sites.
- They remove nucleotides only from the ends of DNA fragments.
- They recognise specific palindromic base-pair sequences.

Choose the answer from the options given below:

- (1) A and B only (2) D and E only
(3) C and D only (4) A and E only

99. Match List-I with List-II.

List-I		List-II	
(A)	Decomposition	(I)	Accumulation of dark coloured amorphous colloidal substance
(B)	Detritus	(II)	Release of inorganic nutrients by the activity of microbes in soil
(C)	Mineralisation	(III)	Breaking down of complex organic matter into inorganic substances
(D)	Humification	(IV)	Dead remains of plants and animals including fecal matter

Choose the **correct** answer from the options given below:

- (1) A-I, B-II, C-III, D-IV
(2) A-IV, B-III, C-I, D-II
(3) A-III, B-IV, C-II, D-I
(4) A-III, B-II, C-I, D-IV

100. In which one of the following, the ovules are **not** enclosed by an ovary wall and remain exposed?

- (1) *Selaginella*
(2) *Funaria*
(3) *Pinus*
(4) *Wolffia*

101. Match List-I with List-II:

List-I (Placentation)		List-II (Example)	
(A)	Marginal	(I)	Mustard
(B)	Axile	(II)	Pea
(C)	Parietal	(III)	Marigold
(D)	Basal	(IV)	Lemon

Choose the **correct** answer from the options given below:

- (1) A-I, B-III, C-II, D-IV
(2) A-IV, B-II, C-I, D-III
(3) A-II, B-IV, C-I, D-III
(4) A-III, B-I, C-IV, D-II

102. In angiosperms, root hairs arise from which one of the following regions of the root?

- (1) The root cap zone
(2) The region of meristematic activity
(3) The region of elongation
(4) The region of maturation

103. Which one of the following is **not** a characteristic of plant cells in the phase of elongation?

- (1) Increased vacuolation
(2) Large conspicuous nuclei
(3) Cell enlargement
(4) New cell wall deposition

104. Which of the following statements are **correct** with reference to a transcription unit?
- A transcription unit in DNA is defined primarily by three regions: promoter, structural gene and terminator.
 - The promoter is said to be located towards the 5'-end of the structural gene.
 - The promoter is a DNA sequence that provides binding site for RNA polymerase.
 - The promoter defines the template and coding strands.
 - The terminator is located towards the 3'-end of the coding strand and it defines the end of the process of transcription.

Choose the **correct** answer from the options given below:

- A, B, C, D and E
 - B, C, D and E only
 - A, C, D and E only
 - A, B, C and D only
105. Alpha-helix is found in which level of protein structure?
- Quaternary structure
 - Tertiary structure
 - Primary structure
 - Secondary structure
106. Which of the following statements are **correct** regarding amino acids ?
- They are substituted methanes.
 - Serine is an aromatic amino acid.
 - Valine is a neutral amino acid.
 - Lysine is an acidic amino acid.

Choose the **correct** answer from the options given below:

- C and D only
 - A and B only
 - A and C only
 - B and C only
107. The main function of bulliform cells in grasses is :
- to make the leaf impermeable to fungal spores.
 - to perform photosynthesis
 - to minimize water loss during water stress.
 - to transport water.

108. Find the **incorrect** statement(s) about photosynthesis from the following:

- The water splitting complex is associated with PS I.
- C₄ plants use the C₃ pathway of CO₂ fixation as the main biosynthetic pathway.
- In C₄ plants, photorespiration does not occur.
- C₃ plants exhibit 'Kranz' anatomy.
- ATP synthesis in chloroplast occurs through chemiosmosis.

Choose the answer from the options given below :

- B only
- A and D only
- B and C only
- B and E only

109. Match List I with List II:

List-I		List-II	
A	Conjunctive tissue	I	Specialised cells in the vicinity of guard cells
B	Casparian strips	II	Endodermal cells rich in starch
C	Subsidiary cells	III	Tissue between xylem and phloem
D	Starch sheath	IV	Endodermal cells with suberin deposition

Choose the **correct** answer from the options given below:

- A-IV, B-III, C-I, D-II
- A-III, B-IV, C-II, D-I
- A-III, B-IV, C-I, D-II
- A-IV, B-III, C-II, D-I

110. Match List-I with List-II:

List-I		List-II	
A	Genetically modified organism	I	<i>Agrobacterium tumefaciens</i>
B	Thermostable DNA polymerase	II	Bt cotton
C	Ti plasmid	III	<i>Thermus aquaticus</i>
D	pBR322	IV	<i>Escherichia coli</i>

Choose the **correct** answer from the options given below:

- A-II, B-I, C-IV, D-III
- A-I, B-IV, C-III, D-II
- A-II, B-III, C-I, D-IV
- A-I, B-II, C-IV, D-III

111. Heterophyllous development in response to environment is an example of which of the following phenomena?

(1) Dedifferentiation
(2) Elasticity
(3) Redifferentiation
(4) Plasticity

112. In racemose inflorescence,

(1) the main axis terminates in a flower
(2) the growth is limited
(3) flowers are borne in an acropetal succession
(4) flowers are solitary

113. Which one of the following disorders is caused by the substitution of Glutamic acid (Glu) by Valine (Val) at the sixth position of the beta globin chain of the haemoglobin molecule?

(1) Haemophilia
(2) Thalassemia
(3) Sickle-cell anaemia
(4) Phenylketonuria

114. Match List-I with List-II:

List-I		List-II	
A.	Incomplete dominance	I.	Human skin colour
B.	Co-dominance	II.	Inheritance of flower colour in <i>Antirrhinum</i> sp.
C.	Pleiotropy	III.	Phenylketonuria disease in humans
D.	Polygenic inheritance	IV.	ABO blood groups

Choose the correct answer from the options given below:

(1) A-II, B-IV, C-III, D-I
(2) A-I, B-III, C-II, D-IV
(3) A-II, B-I, C-III, D-IV
(4) A-I, B-IV, C-III, D-II

115. Arrange the following in the correct developmental sequence related to microsporogenesis:

A. Microspore tetrads
B. Sporogenous tissue
C. Pollen grains
D. Pollen mother cells

Choose the correct answer from the options given below:

(1) D, A, C, B
(2) B, D, C, A
(3) B, D, A, C
(4) A, D, C, B

116. Arrange the following steps of DNA fingerprinting in a correct sequence.

A. Isolation of DNA and its digestion by restriction endonucleases.
B. Hybridisation using a labelled VNTR probe.
C. Transferring of separated DNA fragments to synthetic membranes.
D. Detection of hybridised DNA fragments by autoradiography.
E. Separation of DNA fragments by electrophoresis.

Choose the correct answer from the options given below:

(1) A, E, C, B, D
(2) A, E, B, C, D
(3) A, B, D, C, E
(4) A, D, B, E, C

117. Exploring molecular, genetic and species-level diversity for products of economic importance is called:

(1) Biomagnification
(2) Biofortification
(3) Bioremediation
(4) Bioprospecting

118. Which of the following statements are true with reference to the sex-determination in honeybees?

A. An offspring formed from the union of a sperm and an egg, develops as a female (queen or worker).
B. An unfertilized egg develops as a male by parthenogenesis.
C. A male has half the number of chromosomes than that of a female.
D. Males produce sperms by meiosis.
E. Honeybees have a haplodiploid sex-determination system.

Choose the correct answer from the options given below:

(1) B, C, D and E only
(2) A, B, C and D only
(3) A, B, D and E only
(4) A, B, C and E only

119. Identify the correct sequence of steps in each cycle of Polymerase Chain Reaction:

- (1) Denaturation → Annealing → Extension
- (2) Denaturation → Extension → Annealing
- (3) Extension → Annealing → Denaturation
- (4) Annealing → Denaturation → Extension

120. Which of the following statements are correct with respect to DNA separation, isolation and visualization?

- A. The cutting of DNA is done by molecular scissors.
- B. The DNA fragments separate according to their size in an agarose gel, upon electrophoresis.
- C. The separated DNA fragments can be seen without staining when exposed to UV light.
- D. The separated DNA fragments, when stained with ethidium bromide, can be seen in visible light.

Choose the **correct** answer from the options given below :

- (1) A and D only
- (2) B and D only
- (3) B and C only
- (4) A and B only

121. The main criteria used for Five Kingdom Classification proposed by R.H. Whittaker (1969) included:

- A. Cell structure
- B. Body organization
- C. Presence of flagellum
- D. Reproduction
- E. Phylogenetic relationships

Choose the **correct** answer from the options given below:

- (1) A, B, D and E only
- (2) A, B, C, D and E
- (3) A, B and E only
- (4) B, C and D only

122. Which one of the following is a triploid cell?

- (1) Central cell
- (2) Primary endosperm cell
- (3) Zygote
- (4) Synergid

123. Which of the following statements are correct with reference to packaging of DNA helix?

- A. Histones are organized to form a unit of eight molecules called histone octamer.
- B. Histones are negatively charged basic proteins.
- C. Histones are rich in the basic amino acid residues - lysine and arginine.
- D. The positively charged DNA is wrapped around the histone octamer to form nucleosome.
- E. The packaging of chromatin at higher levels requires an additional set of proteins called non-histone chromosomal proteins.

Choose the **correct** answer from the options given below:

- (1) A, B and D only
- (2) A, C and E only
- (3) C, D and E only
- (4) B, D and E only

124. Which of the following is an *in situ* conservation method?

- (1) Sacred Groves
- (2) Wildlife Safari Parks
- (3) Botanical Gardens
- (4) Seed Banks

125. In the *lac* operon, the *r* gene codes for:

- (1) transacetylase
- (2) the repressor of *lac* operon
- (3) permease
- (4) beta-galactosidase

126. Match List I with List II :

List I (Growth Regulator)		List II (Function/Effect)	
A.	2,4-D	I.	Brewing industry
B.	GAs	II.	Stimulation of stomatal closure
C.	Kinetin	III.	Herbicide
D.	ABA	IV.	Nutrient mobilisation

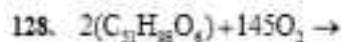
Choose the **correct** answer from the options given below:

- (1) A-IV, B-III, C-II, D-I
- (2) A-I, B-II, C-IV, D-III
- (3) A-III, B-I, C-IV, D-II
- (4) A-I, B-IV, C-III, D-II

127. Arrange the following steps of somatic hybridisation in a correct sequence.
- Digestion of cell walls.
 - Isolation of naked protoplasts.
 - Fusion of protoplasts to get hybrid protoplast.
 - Isolation of single cells from two different varieties of plants.
 - Growing of hybrid protoplast to form a new plant.

Choose the **correct** answer from the options given below:

- E, A, B, C, D
- D, A, B, C, E
- E, B, A, D, C
- D, B, A, E, C



The Respiratory Quotient (RQ) of a biomolecule used for respiration, as per the above equation, would be:

- Less than 0.5
- Between 0.5 and 0.95
- Between 1.25 and 2
- 1.0

129. Since the origin and diversification of life on Earth, there have been five episodes of mass extinction of species. How is the sixth, which is in progress, different from the previous episodes?

- The current species extinction rates are far lower than those in previous episodes.
- The present species extinction rates are 100 to 1000 times faster than in the pre-human times.
- The present net species extinction rate is zero.
- The current species extinction rates are nearly 10 times faster than that in previous episodes.

130. Match List I with List II:

List-I		List-II	
A.	Trypsin	I.	Intercellular ground substance
B.	Morphine	II.	Lectin
C.	Concanavalin A	III.	Enzyme
D.	Collagen	IV.	Alkaloid

Choose the **correct** answer from the options given below:

- A-III, B-IV, C-II, D-I
- A-I, B-II, C-III, D-IV
- A-III, B-II, C-IV, D-I
- A-IV, B-III, C-II, D-I

131. Which one of the following statements is **not** true about the universal rules of binomial nomenclature?

- Both the words in a biological name, when handwritten, are separately underlined and printed in italics.
- The specific epithet in the biological name starts with a small letter.
- The first word in the biological name represents the specific epithet, while the second component denotes the genus.
- Biological names are generally in Latin.

132. The enzyme required for carboxylation in the Calvin cycle is:

- PEP carboxylase
- RuBP carboxylase - oxygenase
- Carboxypeptidase
- Hexokinase

133. Which of the following floral formula is the correct floral formula of Solanaceae family?

- $\oplus \overline{Q}_5 K_{(5)} \overline{C}_{(5)} A_5 \overline{G}_{(2)}$
- $\oplus \overline{Q}_5 K_5 \overline{C}_{(5)} A_5 \overline{G}_{(2)}$
- $\oplus \overline{Q}_5 K_{(5)} C_{(5)} A_5 \overline{G}_{(2)}$
- $\oplus \overline{Q}_5 K_5 C_5 A_5 \overline{G}_{(2)}$

134. Which one of the following types of pollination brings genetically different types of pollen grains to the stigma?

- Geitonogamy
- Autogamy
- Xenogamy
- Cleistogamy

135. Match List I with List II:

List-I (Process)		List-II (Location)	
A.	Glycolysis	I.	Inner mitochondrial membrane
B.	ETS	II.	Mitochondrial matrix
C.	Accumulation of protons	III.	Cytoplasm
D.	Krebs' cycle	IV.	Intermembrane space

Choose the **correct** answer from the options given below:

- A-I, B-IV, C-III, D-II
- A-III, B-I, C-IV, D-II
- A-IV, B-II, C-I, D-III
- A-II, B-III, C-IV, D-I

136. Insertion of a foreign DNA at BamHI site in an *E. coli* cloning vector pBR322 results in the loss of antibiotic resistance towards:

(1) Gentamycin
(2) Ampicillin and tetracycline
(3) Tetracycline
(4) Ampicillin

137. The sixth mutant codon of beta globin gene causing polymerization of Haemoglobin and change in RBC shape is _____.

(1) CAG (2) GUG
(3) AUG (4) GAG

138. Choose the correct statement regarding GIFT to overcome infertility.

(1) Ova collected from a female donor are transferred to the uterus of an infertile female.
(2) It is the transfer of an ovum collected from a donor into the fallopian tube of another female who cannot produce ovum but can provide suitable environment for fertilization and development.
(3) Early embryos with up to 8 blastomeres are transferred to the uterus of an infertile female.
(4) Early embryos with up to 8 blastomeres are transferred into the fallopian tube of an infertile female.

139. Which one of the following is an appropriate example of "sexual deceit"?

(1) Female wasp and fig
(2) Cuckoo and crow
(3) *Ophrys* and bumblebee
(4) Sea anemone and clown fish

140. Evolution of human appears parallel to the progressive development of brain and language skills. As such, the evolution of individual species in the sequence of their appearance is:

(1) *Homo habilis* → *Homo erectus* → *Ramapithecus* → *Neanderthal* → *Homo sapiens*
(2) *Ramapithecus* → *Homo habilis* → *Homo erectus* → *Neanderthal* → *Homo sapiens*
(3) *Homo sapiens* → *Ramapithecus* → *Homo habilis* → *Neanderthal* → *Homo erectus*
(4) *Neanderthal* → *Ramapithecus* → *Homo habilis* → *Homo erectus* → *Homo sapiens*

141. Match List I with List II related to embryonic development at various months of pregnancy:

List I		List II	
A.	The foetus movement starts and hair appears on the head	I.	24 weeks of pregnancy
B.	The foetus develops limbs and digits	II.	20 weeks of pregnancy
C.	The foetus develops external genital organs	III.	8 weeks of pregnancy
D.	The foetus body is covered with fine hair; eyelids separate and eyelashes are formed	IV.	12 weeks of pregnancy

Choose the correct answer from the options given below :

(1) A-III, B-II, C-IV, D-I
(2) A-II, B-IV, C-III, D-I
(3) A-IV, B-II, C-III, D-I
(4) A-II, B-III, C-IV, D-I

142. A group of researchers procured some fish-like animals and upon investigation the following characters were observed :

A. Endoskeleton was made of cartilage.
B. Ectoparasitic, as they were found attached on fish skin with their circular sucking mouth.
C. Paired fins and scales were absent, but 7 pairs of gill slits were present.

Which of the following species of animals did they consider to fit best with these characters?

(1) *Exocoetus* sp. (2) *Branchiostoma* sp.
(3) *Petromyzon* sp. (4) *Scoliodon* sp.

143. Spermatogonia undergo a series of cell divisions to produce sperms. Select the correct statements from the following :

A. Spermatogonia always undergo meiotic cell division.
B. Primary spermatocytes divide mitotically to produce secondary spermatocytes.
C. Secondary spermatocytes, through their second meiotic division, produce haploid spermatids.
D. Spermatids produce spermatozoa through mitosis.
E. Spermatids transform into spermatozoa by spermiogenesis.

Choose the correct answer from the options given below :

(1) C and E only
(2) A, C and E only
(3) B, C and D only
(4) A and E only

144. What is the probability of having children with 'O' blood group, where both mother and father are heterozygous for 'A' and 'B' blood group, respectively?

(1) 50%
(2) 0%
(3) 75%
(4) 25%

145. Arrange the following events occurring in Renin-Angiotensin mechanism in the correct order:

A. Increase in blood pressure and Glomerular filtration rate.
B. Reabsorption of Na^+ and water from distal parts of tubule due to Aldosterone.
C. Fall in Glomerular filtration rate.
D. Vasoconstriction by Angiotensin II and release of Aldosterone.
E. Renin converts Angiotensinogen into Angiotensin I, followed by Angiotensin II.

Choose the correct answer from the options given below:

(1) C, A, B, D, E
(2) A, D, B, E, C
(3) A, C, E, B, D
(4) C, E, D, B, A

146. Match List I with List II.

List I (Respiratory Volume)		List II (Capacity in mL)	
A.	ERV (Expiratory Reserve Volume)	I.	2500 – 3000 mL
B.	RV (Residual Volume)	II.	500 mL
C.	IRV (Inspiratory Reserve Volume)	III.	1000 – 1100 mL
D.	TV (Tidal Volume)	IV.	1100 – 1200 mL

Choose the correct answer from the options given below:

(1) A-III, B-I, C-IV, D-II
(2) A-I, B-III, C-II, D-IV
(3) A-III, B-IV, C-I, D-II
(4) A-I, B-II, C-III, D-IV

147. Match List I with List II.

List I		List II	
A.	Progestasert	I.	Barrier made of rubber used by females
B.	Multiload 375	II.	Oral contraceptive
C.	Diaphragm	III.	Hormone releasing IUD
D.	Saheli	IV.	Copper releasing IUD

Choose the correct answer from the options given below:

(1) A-III, B-IV, C-I, D-II
(2) A-III, B-IV, C-II, D-I
(3) A-IV, B-II, C-I, D-III
(4) A-IV, B-III, C-I, D-II

148. Non-membrane bound cell organelles found in both prokaryotic and eukaryotic cells are _____.

(1) Centrosomes (2) Ribosomes
(3) Lysosomes (4) Mitochondria

149. Ecological pyramids represent the relationship between the organisms at different trophic levels and they are generally inverted for:

(1) Pyramid of energy in pond ecosystem
(2) Pyramid of biomass in sea
(3) Pyramid of number in grassland
(4) Pyramid of biomass in grassland

150. The flightless bird with forelimbs modified as paddle-like structures suited for swimming is known as:

(1) *Struthio* (2) *Pittacula*
(3) *Neophron* (4) *Aptenodytes*

151. Match List-I with List-II.

List-I (Bioactive molecules)		List-II (Importance)	
(A)	Streptokinase	(I)	Immunosuppressive agent
(B)	Statins	(II)	Removal of clots from the blood vessels
(C)	Lipases	(III)	Blood cholesterol-lowering agent
(D)	Cyclosporin A	(IV)	Detergent formulations

Choose the correct answer from the options given below:

(1) A-II, B-III, C-IV, D-I
(2) A-IV, B-III, C-II, D-I
(3) A-II, B-III, C-I, D-IV
(4) A-III, B-II, C-IV, D-I

152. Choose the correct statements regarding cell organelles and their inclusions.
- The endomembrane system includes Golgi complex, endoplasmic reticulum and mitochondria.
 - Rough endoplasmic reticulum bears ribosomes on its surface.
 - Both mitochondria and plastids have circular DNA.
 - A network of microtubules, microfilaments and intermediate filaments present in the cytoplasm is called cytoskeleton.
 - Mitochondrion is a single membrane-bound structure.

Choose the correct answer from the options given below:

- C, D and E only
- A and B only
- A, B and C only
- B, C and D only

153. Select the set of fishes which belong to the class Osteichthyes:

- Devil fish, Cuttlefish and Hagfish
- Starfish, Hagfish and Cuttlefish
- Flying fish, Angel fish and Fighting fish
- Saw fish, Fighting fish and Dog fish

154. In a population of grasshopper species, chromosome number of some members is 23 and some other members possess 24 chromosomes. The 23 and 24 chromosome-bearing members in this species are _____.

- all males
- all females
- females and males, respectively
- males and females, respectively

155. The WBC count of a person's blood sample is 8000/cu.mm. How many eosinophils and lymphocytes would be in the same blood sample approximately?

- 160–240/cu.mm and 1600–2000/cu.mm, respectively
- 100–120/cu.mm and 160–200/cu.mm, respectively
- 300–500/cu.mm and 500–700/cu.mm, respectively
- 300–500/cu.mm and 1200–1500/cu.mm, respectively

156. The toxin proteins isolated from *Bacillus thuringiensis*, coded by which of the following genes would control cotton bollworms and corn borer, respectively?

- cryIAC* and *cryIIAB*
- cryIAC* and *cryIAB*
- cryIAC* and *cryIAB*
- cryIIAB* and *cryIAC*

157. Match List-I with List-II

List-I (Drug)		List-II (Effect)	
(A)	Nicotine	(I)	Causes sense of euphoria and increased energy
(B)	Morphine	(II)	Stimulates adrenal gland to release catecholamines into blood circulation
(C)	Heroin	(III)	Effective sedative and painkiller
(D)	Cocaine	(IV)	A depressant, slows down body function

Choose the correct answer from the options given below:

- A-III, B-II, C-IV, D-I
- A-II, B-III, C-IV, D-I
- A-II, B-III, C-I, D-IV
- A-III, B-II, C-I, D-IV

158. Match List-I with List-II related to muscular/skeletal system:

List-I		List-II	
(A)	Tetany	(I)	Inflammation of joints
(B)	Arthritis	(II)	Autoimmune disorder affecting neuromuscular junction
(C)	Myasthenia gravis	(III)	Wild contraction in muscle due to low Ca^{++} in body fluid
(D)	Muscular dystrophy	(IV)	Progressive degeneration of skeletal muscle

Choose the correct answer from the options given below:

- A-III, B-I, C-II, D-IV
- A-IV, B-III, C-II, D-I
- A-I, B-II, C-III, D-IV
- A-III, B-II, C-I, D-IV

159. In which animal do haploid cells divide mitotically to produce gametes?
 (1) Male honeybees (2) Male grasshoppers
 (3) Male earthworms (4) Male frogs
160. In humans, respiration occurs in the following steps. Arrange these steps in the correct order.
 A. Diffusion of O_2 and CO_2 between blood and tissues
 B. Diffusion of O_2 and CO_2 across alveolar membrane
 C. Pulmonary ventilation by which atmospheric air is drawn in and CO_2 rich alveolar air is released out
 D. Cellular respiration
 E. Transport of gases by the blood.
 Choose the correct answer from the options given below:
 (1) A, B, C, D, E (2) E, A, C, D, B
 (3) C, A, B, E, D (4) C, B, E, A, D
161. Arrange the following cell layers/structure around the female gamete, from outer to inner side:
 A. Zona pellucida
 B. Perivitelline space
 C. Corona radiata
 D. Plasma membrane of ovum
 Choose the correct answer from the options given below:
 (1) C, A, D, B (2) C, A, B, D
 (3) D, B, A, C (4) A, C, B, D
162. The human protein named α -1 antitrypsin, obtained from transgenic animals, is used for the treatment of _____.
 (1) Alzheimer's disease
 (2) Emphysema
 (3) Rheumatoid arthritis
 (4) Cystic fibrosis
163. Select the correct statements regarding cell membrane in eukaryotic cell.
 A. Membrane of human RBCs has approximately 52% protein.
 B. Major phospholipids are arranged in a bilayer.
 C. Extensions of the plasma membrane into the cell form mesosomes.
 D. Tails towards the inner part of lipids are hydrophobic and thus protected from aqueous medium.
 E. Glycocalyx is present on the outer surface of the plasma membrane.
 Choose the correct answer from the options given below:
 (1) A, C and E only (2) B, C and E only
 (3) C, D and E only (4) A, B and D only
164. Male frogs can be distinguished from female frogs due to the presence of:
 A. Bulging eyes
 B. Vocal sacs
 C. Webbed digits in feet
 D. Copulatory pad on first digit of fore limbs
 E. Olive green-coloured skin with dark irregular spots
 Choose the correct answer from the options given below:
 (1) B and D only (2) B and C only
 (3) A and B only (4) C and E only
165. Which of the following equations depicts Verhulst-Pearl logistic population growth?
 (1) $dN/dt = rN(K - N/K)$
 (2) $dN/dt = rN(K + N/K)$
 (3) $dN/dt = rN(K/K - N)$
 (4) $dN/dt = rN(K - N/K)$
166. Choose the correct statements regarding frog's anatomy:
 A. Hepatic portal system is the special venous connection between liver and intestine.
 B. There are twelve pairs of cranial nerves arising from the brain.
 C. The ureters and oviducts open separately into the cloaca in female frogs.
 D. Hind-brain consists of cerebellum, medulla oblongata and optic lobes.
 E. Sinus venosus joins the right atrium of heart.
 Choose the correct answer from the options given below:
 (1) B and D only (2) A, C and E only
 (3) A, B and C only (4) B and C only
167. Select the incorrect statements with reference to Rh grouping.
 A. Erythroblastosis foetalis is a condition observed having foetus with Rh^{++} blood and mother with Rh^{++} blood.
 B. Rh antigen is observed on RBCs in the majority of human beings.
 C. Before blood transfusion, Rh group should also be matched.
 D. Rh incompatibility is observed when a pregnant mother is Rh^{++} and the foetus is Rh^{++} .
 E. Erythroblastosis foetalis can be avoided by administering anti-Rh antibodies to the mother immediately after the delivery of the second child.
 Choose the answer from the options given below:
 (1) A and E only (2) A and B only
 (3) B and C only (4) C and D only

168. Which of the following statements are correct with reference to human endoskeleton ?

- Human skull is monocondylic.
- The joint between any two adjoining vertebrae is a cartilaginous joint.
- In human beings, the number of cervical vertebrae is seven.
- All ribs except the last 2 pairs are bicephalic.
- The occipital bone of skull is articulated with atlas vertebra.

Choose the **correct** answer from the options given below:

- A, B and D only
- B and E only
- B, C and E only
- C, D and E only

169. Match List I with List II:

List I		List II	
A.	Cortisol	I.	Stimulates the formation of alveoli in mammary glands
B.	Aldosterone	II.	Produces anti-inflammatory reactions
C.	Cholecystokinin	III.	Stimulates reabsorption of Na^+ and water from renal tubule
D.	Progesterone	IV.	Stimulates secretion of pancreatic enzymes and bile juice

Choose the **correct** answer from the options given below:

- A-III, B-II, C-IV, D-I
- A-II, B-III, C-IV, D-I
- A-IV, B-II, C-I, D-III
- A-II, B-III, C-I, D-IV

170. The following are the stages of life cycle of *Plasmodium*. Arrange the stages in the proper order.

- The parasites reproduce asexually in RBCs, bursting the cells.
- The parasites reproduce asexually in liver cells, bursting the cells and releasing into blood.
- Gametocytes develop in RBCs.
- Sporozoites reach the liver through the blood.
- Female mosquito injects sporozoites into humans during bite.

Choose the **correct** answer from the options given below:

- A, B, C, D, E
- E, D, B, A, C
- C, A, B, D, E
- E, C, D, B, A

171. Select the **incorrect** statements from the following:

- Digestive system in Platyhelminthes is incomplete.
- Bilateral symmetry is a characteristic feature of adult Echinoderma.
- Pseudocoelom is possessed by Aschelminthes.
- Notochord is persistent throughout life in the class Chondrichthyes.
- Members of class Reptilia maintain a constant body temperature.

Choose the answer from the options given below:

- B and E only
- C and D only
- A and C only
- B and D only

172. The specific receptors for neurotransmitters in a synapse are present on _____

- Post-synaptic membrane
- Pre-synaptic membrane
- Myelin sheath
- Schwann cell

173. Choose the correct statements regarding muscle contraction.

- A motor neuron carries a signal sent by the Central Nervous System (CNS) to the sarcolemma of the muscle fibre.
- The neural signal generates an action potential which causes the release of Ca^{++} into sarcoplasm.
- Increase in Ca^{++} inactivates the actin for breaking cross bridges.
- Actin binds to the myosin head to form a cross bridge.
- Shortening of sarcomere takes place, by pulling actin filaments towards the centre of 'A' band.

Choose the **correct** answer from the options given below:

- A, B, D and E only
- C and D only
- C and E only
- A and B only

174. Which of the following is **not** an example of convergent evolution?

- Eyes of octopuses and mammals
- Fore limbs of whales and bats
- Wings of butterflies and birds
- Flippers of penguins and dolphins

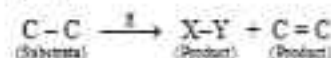
175. The JGA (Juxta Glomerular Apparatus) is a special sensitive region formed by cellular modifications in _____ related to the same nephron.

- (1) Distal convoluted tubule and efferent renal arteriole
- (2) Proximal convoluted tubule and afferent renal arteriole
- (3) Distal convoluted tubule and afferent renal arteriole
- (4) Proximal convoluted tubule and efferent renal arteriole

176. The following reaction depicts the activity of a particular class of enzymes:

X + Y

| |



Identify the enzyme class 'E' from the following options:

- (1) Ligases (2) Lyases
- (3) Isomerases (4) Transferases

177. Match List I with List II:

List I		List II	
A.	Molluscs	I.	Pulmonary respiration only
B.	Reptiles	II.	Branchial respiration
C.	Adult amphibians	III.	Cellular respiration
D.	Amoeba	IV.	Pulmonary and Cutaneous respiration

Choose the correct answer from the options given below:

- (1) A-III, B-II, C-I, D-IV
- (2) A-II, B-I, C-IV, D-III
- (3) A-II, B-I, C-III, D-IV
- (4) A-I, B-II, C-IV, D-III

178. What is the reason behind production of large holes in 'Swiss Cheese'?

- (1) The production of large amount of CO_2 by *Clostridium butylicum*
- (2) The production of large amount of CO_2 and H_2 by *Trichoderma polysporum*
- (3) The production of large amount of CO_2 and H_2 by lactic acid bacteria called *Lactobacillus*
- (4) The production of large amount of CO_2 by *Propionibacterium sharmanii*

179. Match List I with List II with respect to chronology of evolution of life forms:

List I		List II	
A.	About 65 mya	I.	Jawless fish probably evolved
B.	About 500 mya	II.	The dinosaurs suddenly disappeared from the earth
C.	About 350 mya	III.	Seaweeds and few plants probably existed
D.	About 320 mya	IV.	Invertebrates were formed and became active

Choose the correct answer from the options given below:

- (1) A-II, B-IV, C-I, D-III
- (2) A-II, B-IV, C-III, D-I
- (3) A-I, B-II, C-III, D-IV
- (4) A-III, B-IV, C-I, D-II

180. Choose the correct statements regarding population interactions between two species.

- A. In both parasitism and commensalism, only one species benefits and the other species is harmed.
- B. Both species benefit in mutualism.
- C. Both species benefit in commensalism.
- D. In parasitism, only one species benefits and the other species is harmed.
- E. In amensalism, one species is harmed and the other is unaffected.

Choose the correct answer from the options given below:

- (1) A and D only
- (2) A and B only
- (3) B and E only
- (4) B, D and E only



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ANSWER KEY

1. (1)	46. (4)	91. (2)	136. (3)
2. (1)	47. (2)	92. (1)	137. (2)
3. (2)	48. (3)	93. (1)	138. (2)
4. (3)	49. (3)	94. (2)	139. (3)
5. (1)	50. (1)	95. (2)	140. (2)
6. (3)	51. (2)	96. (2)	141. (4)
7. (3)	52. (4)	97. (1)	142. (3)
8. (1)	53. (4)	98. (3)	143. (1)
9. (1)	54. (2)	99. (3)	144. (4)
10. (1)	55. (1)	100. (3)	145. (4)
11. (1)	56. (3)	101. (3)	146. (3)
12. (1)	57. (1)	102. (4)	147. (1)
13. (1)	58. (3)	103. (2)	148. (2)
14. (3)	59. (3)	104. (1)	149. (2)
15. (1)	60. (1)	105. (4)	150. (4)
16. (4)	61. (2)	106. (3)	151. (1)
17. (4)	62. (3)	107. (3)	152. (4)
18. (4)	63. (4)	108. (2)	153. (3)
19. (2)	64. (3)	109. (3)	154. (4)
20. (1)	65. (1)	110. (3)	155. (1)
21. (2)	66. (2)	111. (4)	156. (3)
22. (4)	67. (2)	112. (3)	157. (2)
23. (1)	68. (3)	113. (3)	158. (1)
24. (2)	69. (1)	114. (1)	159. (1)
25. (3)	70. (2)	115. (3)	160. (4)
26. (2)	71. (1)	116. (1)	161. (2)
27. (4)	72. (2)	117. (4)	162. (2)
28. (3)	73. (4)	118. (4)	163. (4)
29. (2)	74. (1)	119. (1)	164. (1)
30. (4)	75. (3)	120. (4)	165. (1)
31. (2)	76. (2)	121. (1)	166. (2)
32. (4)	77. (1)	122. (2)	167. (1)
33. (1)	78. (3)	123. (2)	168. (3)
34. (1)	79. (2)	124. (1)	169. (2)
35. (4)	80. (1)	125. (4)	170. (2)
36. (1)	81. (3)	126. (3)	171. (1)
37. (1)	82. (4)	127. (2)	172. (1)
38. (4)	83. (1)	128. (2)	173. (1)
39. (4)	84. (4)	129. (2)	174. (2)
40. (4)	85. (2)	130. (1)	175. (3)
41. (3)	86. (2)	131. (3)	176. (2)
42. (1)	87. (3)	132. (2)	177. (2)
43. (4)	88. (4)	133. (1)	178. (4)
44. (2)	89. (4)	134. (3)	179. (1)
45. (1)	90. (3)	135. (2)	180. (4)

Hints & Solutions

Q1 Text Solution:

Using NCERT formulae: $B = \frac{\mu_0 NI}{2R}$ and magnetic moment ($m = NI A$).

$$\left[I = \frac{2RB}{\mu_0 N} = \frac{2(0.05)(3.14 \times 10^{-3})}{(4\pi \times 10^{-7})(100)} = 2.5 \text{ A} \right]$$

$$\left[m = NI \pi R^2 = 100(2.5)\pi(0.05)^2 \right]$$

$$\approx 2 \text{ A m}^2]$$

Q2 Text Solution:

A. $E = h\nu \rightarrow$ Energy of photon (IV)

B. Diffraction and interference $\rightarrow$ Wave nature of light (III)

C. $\lambda = \frac{h}{p} \rightarrow$ de Broglie wavelength (I)

D. Compton effect $\rightarrow$ Particle nature of light (II)

NCERT states $E = h\nu$ for photon energy, Compton effect confirms particle-like behaviour, and $\lambda = h/p$ is the de Broglie wavelength.

Q3 Text Solution:

Left rail is at higher potential (+10 V).

So only the diodes allowing current left to right conduct.

Conducting branches: (4Ω) and (2Ω)

Blocked branches: (3Ω) and (5Ω)

$$R_{eq} = \frac{4 \times 2}{4+2} = \frac{4}{3} \Omega$$

$$I = \frac{V}{R_{eq}} = \frac{10}{4/3} = \frac{15}{2} \text{ A}$$

So, option (2).

Q4 Text Solution:

Time taken:

$$6 \text{ min } 40 \text{ s} = 6 \times 60 + 40 = 400 \text{ s}$$

Speed of light is taken as unity:

$$c = 1$$

$$\text{Distance} = ct = 1 \times 400 = 400$$

So, option (3).

Q5 Text Solution:

For a ball thrown vertically upward, acceleration is constant downward ($-g$).

Taking upward as positive:

$$v = u - gt$$

So velocity starts positive, decreases linearly, becomes zero at the highest point, then becomes

negative while falling.

Hence the correct (v)-(t) graph is **C** only.

Q6 Text Solution:

Given: 20 VSD = 16 MSD

$$1 \text{ VSD} = \frac{16}{20} \text{ MSD} = 0.8 \text{ mm}$$

$$\text{Least count} = 1 \text{ MSD} - 1 \text{ VSD} = 1 - 0.8$$

$$= 0.2 \text{ mm}$$

$$0.2 \text{ mm} = 0.02 \text{ cm}$$

Q7 Text Solution:

For series LCR resonance:

$$f = \frac{1}{2\pi\sqrt{LC}}$$

$$L = 1 \text{ mH} = 10^{-3} \text{ H}, C = 0.1 \mu\text{F}$$

$$= 10^{-7} \text{ F}$$

$$f = \frac{1}{2\pi\sqrt{10^{-3} \times 10^{-7}}} = \frac{1}{2\pi \times 10^{-5}} \approx 1.59 \times 10^4 \text{ Hz}$$

$$= 15.9 \text{ kHz}$$

Q8 Text Solution:

For a uniformly distributed current in a solid wire:

Inside ($r < a$):

$$B = \frac{\mu_0 I r}{2\pi a^2}$$

So, (B or r) linear increase.

Outside ($r > a$):

$$B = \frac{\mu_0 I}{2\pi r}$$

Q9 Text Solution:

Total wire resistance = 4Ω , so each side of square:

$$R_{side} = 1\Omega$$

Battery is across opposite corners (A) and (C). By symmetry, (B) and (D) are at the same potential, so no current flows through the 2Ω diagonal resistor.

Two equal paths:

$$A - B - C = 1 + 1 = 2\Omega$$

$$A - D - C = 1 + 1 = 2\Omega$$

These are in parallel:

$$R_{eq} = \frac{2 \times 2}{2+2} = 1\Omega$$

$$I = \frac{V}{R} = \frac{2}{1} = 2 \text{ A}$$

Q10 Text Solution:

$$R = R_0 A^{1/3},$$



so nuclear volume = $\frac{4}{3}\pi R_0^3 A$.

$$\rho = \frac{m}{V} = \frac{m}{\frac{4}{3}\pi R_0^3 A}$$

$$A = \frac{m}{\rho \cdot \frac{4}{3}\pi R_0^3}$$

$$A = \frac{19.926 \times 10^{-27}}{(2.39 \times 10^{17}) \left(\frac{226}{4}\right) (1.2 \times 10^{-15})^3} \approx 12$$

Q11 Text Solution:

For rectangular loop moving out of magnetic field:

$$\epsilon = Blv$$

Here, velocity is normal to the shorter side, so

$$l = 3 \text{ cm} = 0.03 \text{ m}$$

$$\epsilon = 0.3 \times 0.03 \times 0.02 = 1.8 \times 10^{-4} \text{ V}$$

Q12 Text Solution:

For shunt:

$$S = \frac{I_g G}{I - I_g}$$

$$I_g = 1 \text{ mA} = 10^{-3} \text{ A}, G = 100 \Omega, I = 10 \text{ A}$$

$$S = \frac{10^{-3} \times 100}{10 - 10^{-3}} = \frac{0.1}{9.999} \approx 0.01 \Omega$$

Q13 Text Solution:

In YDSE, for equal intensities:

$$I = I_{\max} \cos^2 \left(\frac{\phi}{2} \right)$$

At path difference (λ), intensity is maximum, so:

$$I_{\max} = K$$

For path difference ($\lambda/3$):

$$\phi = \frac{2\pi}{\lambda} \cdot \frac{\lambda}{3} = \frac{2\pi}{3}$$

$$I = K \cos^2 \left(\frac{\pi}{3} \right) = K \left(\frac{1}{2} \right)^2 = \frac{K}{4}$$

Q14 Text Solution:

Resultant force:

$$F = \sqrt{8^2 + 6^2} = 10 \text{ N}$$

$$a = \frac{F}{m} = \frac{10}{5} = 2 \text{ m s}^{-2}$$

Direction with 8 N force:

$$\tan \theta = \frac{6}{8} = \frac{3}{4}$$

Q15 Text Solution:

(C_1, C_2, C_3, C_4) are in series:

$$C_s = \frac{10}{4} = 2.5 \mu\text{F}$$

This is parallel with $C_5 = 2.5 \mu\text{F}$:

$$C_{eq} = 2.5 + 2.5 = 5 \mu\text{F}$$

Each parallel branch gets (50 V).

$$Q_{\text{series branch}} = 2.5 \times 50 = 125 \mu\text{C}$$

In series, charge is same on each capacitor, so (C_1) to (C_4) each have (125 μC). Also, $Q_5 = 2.5 \times 50 = 125 \mu\text{C}$

Q16 Text Solution:

A metre bridge is a Wheatstone bridge. Interchanging the cell and galvanometer does not change the balance condition.

So, on two sides of the balance point, galvanometer deflects in opposite directions, and at balance:

$$G = 0 \text{ (Deflection in galvanometer is zero)}$$

Q17 Text Solution:

$$P = \frac{W}{t} = \frac{mgh}{t}$$

$$P = \frac{1000 \times 9.8 \times 20}{10} = 19600 \text{ W}$$

$$19600 \text{ W} = 19.6 \text{ kW}$$

Q18 Text Solution:

A. Young's modulus:

$$Y = \frac{FL}{A\Delta L}$$

B. Compressibility:

$$K = -\frac{1}{\Delta P} \left(\frac{\Delta V}{V} \right)$$

C. Bulk modulus:

$$B = -P \left(\frac{V}{\Delta V} \right)$$

D. Poisson's ratio:

$$\sigma = \frac{\Delta d/d}{\Delta L/L} = \frac{\Delta d}{\Delta L} \left(\frac{L}{d} \right)$$

Q19 Text Solution:

For a **concave lens**, a ray parallel to the principal axis diverges after refraction and appears to come from the **same-side focus**.

Q20 Text Solution:

Here (m) is linear mass density, so total mass:

$$M = mL$$

$$L = 2\pi R \Rightarrow R = \frac{L}{2\pi}$$

Moment of inertia of ring about diameter:

$$I_C = \frac{MR^2}{2}$$

Axis (yy') is tangent and parallel to diameter, so by parallel axis theorem:

$$I = I_C + MR^2 = \frac{3}{2}MR^2$$

$$I = \frac{3}{2} \left(mL \right) \left(\frac{L}{2\pi} \right)^2 = \frac{3mL^3}{8\pi^2}$$

Q21 Text Solution:



$$\rho = \frac{m}{s^3}$$

$$a = 9.0 \text{ cm} = 0.090 \text{ m}$$

$$\rho = \frac{5.580}{(0.090)^3} = \frac{5.580}{0.000729} = 7654 \text{ kg m}^{-3}$$

$$\rho = 7.654 \times 10^3 \text{ kg m}^{-3}$$

Here, least significant figures are in $(9.0) \text{ cm} = 2$ significant figures, so:

$$X = 7.7$$

Q22 Text Solution:

Given:

$$y = 2 \cos 2\pi(10t - 0.0080x + 0.35)$$

Phase difference for distance Δx :

$$\Delta\phi = 2\pi(0.0080)\Delta x$$

$$\Delta x = 0.5 \text{ m} = 50 \text{ cm}$$

$$\Delta\phi = 2\pi(0.0080)(50) = 0.8\pi \text{ rad}$$

Q23 Text Solution:

For an equilateral prism:

$$A = 60^\circ$$

Since refracted ray (QR) is parallel to base, it makes 30° with the normal at each face:

$$r_1 = r_2 = 30^\circ$$

By reversibility/symmetry:

$$e = i = 50^\circ$$

Deviation:

$$\delta = i + e - A = 50^\circ + 50^\circ - 60^\circ = 40^\circ$$

Q24 Text Solution:

The diode is **reverse biased during the positive half-cycle**, so the full positive half of input appears across the diode.

During the negative half-cycle, the diode is **forward biased**, so for an ideal diode:

$$V_D = 0$$

Hence (V_D) is a **positive half-wave only**

Q25 Text Solution:

For a simple pendulum in SHM:

$$\text{K.E.} = \frac{1}{2}mv^2$$

Velocity is zero at extreme positions, so K.E. is zero at:

$$t = 0, \frac{T}{2}, T$$

K.E. is always **positive** and maximum at mean position.

So, the correct graph is **option (3)**.

Q26 Text Solution:

Terminal voltage:

$$V = E - Ir$$

$$V = 12 - (0.6)(2) = 12 - 1.2 = 10.8 \text{ V}$$

Q27 Text Solution:

Work done:

$$W = \Delta U = GMm \left(\frac{1}{R} - \frac{1}{2R} \right)$$

$$W = \frac{GMm}{2R}$$

Since,

$$g = \frac{GM}{R^2} \Rightarrow GM = gR^2$$

$$W = \frac{mgR}{2}$$

Q28 Text Solution:

Using first law of thermodynamics:

$$\frac{dQ}{dt} = \frac{dU}{dt} + \frac{dW}{dt}$$

$$100 = \frac{dU}{dt} + 75$$

$$\frac{dU}{dt} = 25 \text{ W}$$

Q29 Text Solution:

For a heater, resistance remains constant:

$$R = \frac{V^2}{P} = \frac{220^2}{400} = 121 \Omega$$

At (200 V) :

$$P' = \frac{V'^2}{R} = \frac{200^2}{121} \approx 331 \text{ W}$$

Q30 Text Solution:

Distance fallen:

$$s = \frac{1}{2}gt^2$$

So greater reaction time means greater distance.

Times:

$$B = 0.22 > E = 0.21 > A = 0.20 >$$

$$D = 0.19 > C = 0.18$$

Q31 Text Solution:

$$U_i = \frac{1}{2}CV^2$$

$$\text{Here } C = 200 \text{ pF}, V = 100 \text{ V}$$

When a charged capacitor is connected to an uncharged capacitor, charge is redistributed until both reach a common potential.

Since the capacitors are identical, the charge will split **exactly in half** between them.

So final voltage across each:

$$V_f = \frac{V}{2}$$

$$U_f = \frac{1}{2} \left(C \right) \left(\frac{V}{2} \right)^2 + \frac{1}{2} \left(C \right) \left(\frac{V}{2} \right)^2 = \frac{1}{4} CV^2$$

Energy lost:

$$\Delta U = \frac{1}{2} CV^2 - \frac{1}{4} CV^2 = \frac{1}{4} CV^2$$



$$= \frac{1}{4} (200 \times 10^{-12}) (100)^2 = 0.5 \times 10^{-6} \text{ J}$$

Q32 Text Solution:

Time period:

$$T = \frac{60}{30} = 2 \text{ s}$$

For simple pendulum:

$$T = 2\pi \sqrt{\frac{L}{g}}$$

$$L = \frac{gT^2}{4\pi^2} = \frac{9.8 \times 4}{4 \times 9.8} = 1 \text{ m}$$

So, length = 1 m.

Q33 Text Solution:

For AC starting from zero:

$$i = i_0 \sin \omega t$$

Peak value occurs when:

$$\sin \omega t = 1 \Rightarrow \omega t = \frac{\pi}{2}$$

$$\frac{2\pi}{T} t = \frac{\pi}{2}$$

So time taken is one-fourth of time period:

$$t = \frac{T}{4} = \frac{1}{4f}$$

$$t = \frac{1}{4 \times 60} = \frac{1}{240} \text{ s}$$

Q34 Text Solution:

A is true: In interference/diffraction, energy is only redistributed. Dark fringes have less energy and bright fringes have more energy, so conservation of energy is obeyed.

B is false: Interference and diffraction are wave phenomena, not exclusive to light. Sound waves, water waves, etc. also show them.

Q35 Text Solution:

For the box to remain stationary relative to trolley, static friction must provide acceleration:

$$\mu_s mg \leq f_{s, \max} = \mu_s mg$$

$$a_{\max} = \mu_s g = 0.12 \times 10 = 1.2 \text{ m s}^{-2}$$

Mass cancels out.

Q36 Text Solution:

Total mechanical energy:

$$E = K + U = 0.02 \text{ J}$$

At equilibrium position, potential energy is minimum/taken zero, so:

$$K = 0.02 \text{ J}$$

$$\frac{1}{2}mv^2 = 0.02$$

Mass:

$$m = 20 \text{ g} = 0.02 \text{ kg}$$

$$v = \sqrt{\frac{2E}{m}} = \sqrt{\frac{2 \times 0.02}{0.02}} = \sqrt{2} \approx 1.41 \text{ m/s}$$

Q37 Text Solution:

Nuclear radius:

$$R = R_0 A^{1/3}$$

So nuclear volume:

$$V \propto R^3 \propto A$$

Hence **A is false, B is true.**

Mass defect is the difference between the mass of a nucleus and the total mass of its constituent protons and neutrons. Hence **C is false, D is true.**

Q38 Text Solution:

Assuming uniform angular acceleration:

$$600 \text{ rpm} = 10 \text{ rps}, \quad 1200 \text{ rpm} = 20 \text{ rps}$$

Average frequency:

$$f_{\text{avg}} = \frac{10+20}{2} = 15 \text{ rev/s}$$

Number of revolutions in 10 s:

$$N = f_{\text{avg}} t = 15 \times 10 = 150$$

Q39 Text Solution:

At depth h ,

$$P_{\text{absolute}} = P_{\text{atm}} + \rho gh$$

Given:

$$100 \text{ atm} = 1 \text{ atm} + \rho gh$$

$$\rho gh = 99 \text{ atm} = 99 \times 10^5 \text{ Pa}$$



$$h = \frac{6.6 \times 10^{-34}}{6.6 \times 10^{-37}} = 990 \text{ m}$$

So, maximum depth = 990 m.

Q40 Text Solution:

- **Microwave** → Klystron valve or magnetron valve
- **Visible light** → Electrons jump from higher to lower energy levels
- **Gamma rays** → Radioactive decay of nucleus
- **Infra-red rays** → Vibrations of atoms and molecules

Q41 Text Solution:

- **A true:** In electrostatic equilibrium, electric field inside a conductor is zero.
- **B false:** Electric field just outside surface depends on surface charge density:

$$E = \frac{\sigma}{\epsilon_0}$$

- **C true:** Excess charge resides only on the surface of a charged conductor.
- **D true:** Electric field at conductor surface is normal to the surface.
- **E false:** Potential inside a charged conductor is constant, not necessarily zero.

Q42 Text Solution:

Photoelectric effect occurs only if:

$$\frac{hc}{\lambda} \geq \phi$$

Threshold wavelength:

$$\lambda_0 = \frac{hc}{\phi}$$

$$\phi = 6.6 \text{ eV} = 6.6 \times 1.6 \times 10^{-19} \text{ J}$$

$$\lambda_0 = \frac{6.6 \times 10^{-34} \times 3 \times 10^8}{6.6 \times 1.6 \times 10^{-19}} = 1.875 \times 10^{-7} \text{ m}$$

= 187.5 nm
Wavelength greater than (187.5 nm) will not produce photoelectric effect.

$$200 \text{ nm} > 187.5 \text{ nm}$$

Q43 Text Solution:

For hydrogen atom,

$$E = -\frac{ke^2}{2r}$$

So,

$$r = \frac{ke^2}{2|E|}$$

$$E = 3.4 \text{ eV} = 3.4 \times 1.6 \times 10^{-19} \text{ J}$$

$$r = \frac{9 \times 10^9 (1.6 \times 10^{-19})^2}{2 \times 3.4 \times 1.6 \times 10^{-19}} \approx 2.1 \times 10^{-10} \text{ m}$$

Q44 Text Solution:

A is true: In forward bias, after threshold/cut-in voltage, diode current increases significantly.

B is false: This current is **forward current**, not reverse saturation current. Reverse saturation current is the small current in reverse bias.

Q45 Text Solution:

For a gas molecule:

$$v_{\text{rms}} = \sqrt{\frac{3RT}{M}}$$

At same temperature,

$$\frac{v_{\text{rms}}}{v_{\text{rms}}} = \sqrt{\frac{M_{\text{O}_2}}{M_{\text{Ar}}}}$$

$$= \sqrt{\frac{10}{40}} = \sqrt{\frac{1}{4}} = \frac{\sqrt{1}}{2}$$

The given mass ratio 2:1 is irrelevant here.

Q46 Text Solution:

A-II: Cumene → phenol by oxidation with O_2 followed by acid hydrolysis.

B-III: $\text{CH}_3\text{COOH} \rightarrow \text{CH}_3\text{CH}_2\text{OH}$ via esterification with $\text{CH}_3\text{OH}/\text{H}^+$, then hydrogenation.

C-IV: 1-propanol → 2-propanol by dehydration to propene, then acid-catalysed hydration.

D-I: Benzene → phenol by sulphonation with oleum, NaOH fusion, then acidification.

So, A-II, B-III, C-IV, D-I.

Q47 Text Solution:

Sequence:



$C_2H_6 \rightarrow C_2H_5Cl$ by chlorination

$C_2H_5Cl + NH_3 \rightarrow C_2H_5NH_2$ by ammonolysis

$C_2H_5NH_2 + HNO_2 \rightarrow C_2H_5OH + N_2 + H_2O$

Primary aliphatic amines react with nitrous acid to give alcohols with evolution of N_2 .

Q48 Text Solution:

For



$$K = [BiO^+][OH^-]$$

Let solubility = s , so:

$$s^2 = 4 \times 10^{-10}$$

$$s = [OH^-] = 2 \times 10^{-5}$$

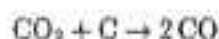
$$pOH = -\log(2 \times 10^{-5}) = 5 - 0.301 = 4.699$$

$$pH = 14 - 4.699 = 9.301$$

So, $pH = 9.301$

Q49 Text Solution:

Reaction with hot coke:



Let $x \text{ dm}^3$ of CO_2 react.

Remaining $CO_2 = 1 - x$

CO formed = $2x$

$$(1 - x) + 2x = 1.4$$

$$1 + x = 1.4 \Rightarrow x = 0.4$$

So,

$$CO = 0.8 \text{ dm}^3, \quad CO_2 = 0.6 \text{ dm}^3$$

Q50 Text Solution:

Use l -values:

$$l = 0 \rightarrow s, \quad l = 1 \rightarrow p, \quad l = 2 \rightarrow d, \quad l = 3 \rightarrow f$$

A: $n=2, l=1 \rightarrow 2p \rightarrow II$

B: $n=4, l=0 \rightarrow 4s \rightarrow III$

C: $n=5, l=3 \rightarrow 5f \rightarrow IV$

D: $n=3, l=2 \rightarrow 3d \rightarrow I$

So, A-II, B-III, C-IV, D-I.

Q51 Text Solution:

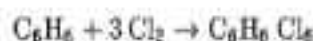
With anhyd. $AlCl_3$, dark/cold, benzene

undergoes electrophilic substitution with excess chlorine:



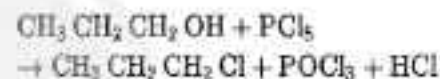
So, X = hexachlorobenzene, has 6 Cl atoms.

With UV, 500 K, benzene undergoes addition:



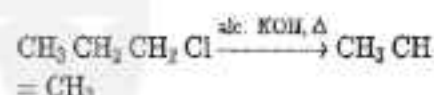
So, Y = benzene hexachloride/BHC, has 6 Cl atoms.

Q52 Text Solution:

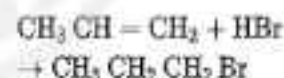


So, X = $POCl_3$.

Then:



Propene reacts with HBr in presence of benzoyl peroxide by anti-Markovnikov addition:



So, Z = $CH_3CH_2CH_2Br$.

Answer: X = $POCl_3$, Z = $CH_3CH_2CH_2Br$.

Q53 Text Solution:

A. $V_2O_5 \rightarrow$ Preparation of H_2SO_4 from SO_2

B. Fe $\rightarrow$ Preparation of ammonia from N_2/H_2 mixture

C. $PdCl_2 \rightarrow$ Oxidation to ethanal

D. Ni complex $\rightarrow$ Polymerisation of alkynes

Q54 Text Solution:



Central Cl has:

- 3 bond pairs
- 2 lone pairs

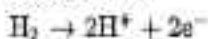


So VSEPR type is:



Electron-pair geometry is trigonal bipyramidal, but molecular shape is T-shaped.

Q55 Text Solution:



$$E_{\text{cell}} = E^{\circ}_{\text{cell}} - \frac{0.059}{2} \log \frac{(2 \times 10^{-4})^2}{2}$$

$$= 0 - \frac{0.059}{2} \log \frac{4 \times 10^{-4}}{2}$$

$$= -\frac{0.059}{2} [\log 2 - 4]$$

$$= -0.0295 [0.3010 - 4]$$

$$= +0.109$$

Q56 Text Solution:

Unit of rate constant:

$$k = (\text{mol L}^{-1})^{1-n} \text{s}^{-1}$$

So:

A. Zero order $\rightarrow \text{mol L}^{-1} \text{s}^{-1} \rightarrow \text{IV}$

B. First order $\rightarrow \text{s}^{-1} \rightarrow \text{III}$

C. Second order $\rightarrow \text{L mol}^{-1} \text{s}^{-1} \rightarrow \text{I}$

D. Third order $\rightarrow \text{L}^2 \text{mol}^{-2} \text{s}^{-1} \rightarrow \text{II}$

Q57 Text Solution:

For Ti^{2+} :



Number of unpaired electrons:

$$n = 2$$

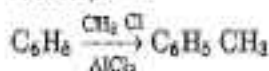
Spin-only magnetic moment:

$$\mu = \sqrt{n(n+2)}$$

$$\mu = \sqrt{2(2+2)} = \sqrt{8} = 2.84 \text{ BM}$$

Q58 Text Solution:

First step:



This forms **toluene**.

On nitration, toluene gives mainly:

o-nitrotoluene and *p*-nitrotoluene

These can be separated by **fractional distillation** due to difference in their boiling points.

Q59 Text Solution:

Power converted into light:

$$150 \times \frac{8}{100} = 12 \text{ J s}^{-1}$$

Energy of one photon:

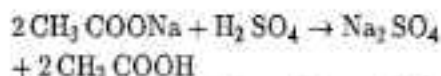
$$4.42 \times 10^{-19} \text{ J}$$

Number of photons per second:

$$\frac{12}{4.42 \times 10^{-19}} = 2.71 \times 10^{19}$$

Q60 Text Solution:

Dilute H_2SO_4 reacts with acetate salt to liberate **acetic acid vapours**, which have a **vinegar smell** and turn blue litmus red.



This is the characteristic preliminary test for acetate ion.

Q61 Text Solution:

Nitriles ($R-C \equiv N$) reduce to primary amines ($R-CH_2NH_2$) with:

A. $(LiAlH_4/H_2O)$ —yes

C. (H_2/Ni) —yes

D. $(Na(Hg)/C_2H_5OH)$ —yes

B. $(Sn + HCl)$ is not the standard reduction of nitrile to primary amine; it is associated with other reductions/Stephen-type reduction.

E. $(Br_2/aq. NaOH)$ is Hofmann bromamide reaction for amides, not nitriles.

Q62 Text Solution:

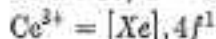
Oxygen can also show:

-1 in peroxides, $-\frac{1}{2}$ in superoxides, and positive oxidation states in compounds with fluorine, e.g. OF_2 .

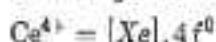
Other statements are correct.

Q63 Text Solution:

Cerium: Ce, $Z = 58$



On losing one more electron:



So, Ce^{4+} is stable because it attains an empty $4f^0$ configuration.

Q64 Text Solution:

In Lassaigne's test, covalently bonded elements in organic compounds like N, S, X are fused with sodium and converted into ionic sodium salts, e.g.



So detection becomes easier in aqueous extract.

Q65 Text Solution:

Urea: NH_2CONH_2

Each urea molecule has **4 H atoms**.

Moles of urea:

$$\frac{5.4}{60} = 0.09 \text{ mol}$$

Number of H atoms:

$$0.09 \times 6.022 \times 10^{23} \times 4 = 2.168 \times 10^{23}$$

Q66 Text Solution:

$\text{CH}_3\text{OCH}_2\text{CH}_2\text{CH}_3$

and

$\text{CH}_3\text{CH}_2\text{OCH}_2\text{CH}_3$

Both are ethers with the same molecular formula

$\text{C}_4\text{H}_{10}\text{O}$, but different alkyl groups on the two sides of oxygen.

So they are **metamers**.

Q67 Text Solution:

Nitrogen is a **2nd-period element** and has **no vacant d-orbitals**, so it cannot form a $d\pi - p\pi$ bond with oxygen.

Nitrogen forms $p\pi - p\pi$ multiple bonding, e.g. in:



Q68 Text Solution:

Phenolphthalein is **colourless in acidic solution** and **pink in alkaline solution**.

In oxalic acid vs NaOH titration, near equivalence point the medium becomes slightly alkaline, so the endpoint is:

colourless $\rightarrow$ pink

Q69 Text Solution:

A. C_2H_4 : $5\sigma, 1\pi$

B. C_2H_2 : $3\sigma, 2\pi$

C. CH_4 : 4σ

D. NH_3 : 3σ , one lone pair

Q70 Text Solution:

Heat absorbed by system:

$$q = +500 \text{ J}$$

Work done by system:

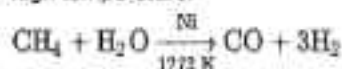
$$w = -200 \text{ J}$$

So,

$$\Delta U = q + w = 500 - 200 = 300 \text{ J}$$

Q71 Text Solution:

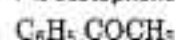
Methane reacts with steam over Ni catalyst at high temperature:



This is **steam reforming of methane**.

Q72 Text Solution:

P is **acetophenone**:



It gives 2,4-DNP test because it has a carbonyl group, and it does not reduce Fehling's reagent because it is a ketone.

On drastic oxidation with chromic acid:



Q is **benzoic acid**, which gives effervescence with NaHCO_3 due to CO_2 evolution.

Q73 Text Solution:

For copper deposition:



$$Q = It = 1.5 \times 10 \times 60 = 900 \text{ C}$$

$$m = \frac{MQ}{nF} = \frac{63.5 \times 900}{2 \times 96487} \approx 0.2938 \text{ g}$$

Q74 Text Solution:

Phthalein dye test is used to identify **phenolic - OH group**.

Phenol reacts with phthalic anhydride in presence of conc. H_2SO_4 to form phenolphthalein-type dye, which gives pink colour in alkaline medium.

Q75 Text Solution:

DNA possesses a **double-stranded helical structure** and contains **thymine** as one of its four nitrogenous bases.

RNA is generally **single-stranded** and contains **uracil instead of thymine**.

Q76 Text Solution:

A. Correct:

$$m = \frac{2.5/60}{75/1000} = 0.556 \text{ m}$$

B. Correct:

$$M = \frac{3/40}{0.450} = 0.278 \text{ M}$$

C. Correct: Cold water dissolves more O_2 , so aquatic species are more comfortable.



D. Incorrect: Gas solubility increases with increase in pressure.

E. Incorrect:

$$\alpha_B = \frac{n_B}{n_A + n_B} \text{ not } \frac{n_A}{n_A + n_B}$$

Q77 Text Solution:

Chloroform and acetone show **negative deviation** from Raoult's law because new **stronger intermolecular attraction** forms between them.



This is hydrogen bonding between the acidic H of chloroform and oxygen of acetone.

Q78 Text Solution:

Given:

$$K_b(X^-) = 10^{-10}$$

So,

$$pK_b = 10$$

For conjugate acid-base pair:

$$pK_a + pK_b = 14$$

$$pK_a = 14 - 10 = 4$$

$$\text{Since } [X^-] = [HX],$$

$$pH = pK_a = 4$$

Q79 Text Solution:

Among Mg , Mg^{2+} , Al , Al^{3+} :

Largest = Mg

Smallest = Al^{3+}

Reason: atomic size decreases from Mg to Al across the period, and cations are smaller than neutral atoms. Also, Mg^{2+} and Al^{3+} are isoelectronic, but Al^{3+} has higher nuclear charge, so it is smaller.

Q80 Text Solution:

$\text{P} < \text{Si} < \text{Be} < \text{Mg} < \text{Na}$

Metallic character **decreases left to right** across a period and **increases down a group**.

So among these, P is least metallic and Na is most metallic.

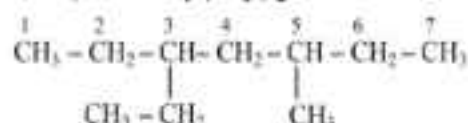
Q81 Text Solution:

Longest chain has 7 carbons, so parent hydrocarbon is **heptane**.

Substituents:

3-ethyl, 5-methyl

Since locants are (3,5) from both directions, numbering is chosen so the substituent written first alphabetically (**ethyl**) gets the lower number.



So correct name is:

3-ethyl-5-methylheptane

Q82 Text Solution:

A. $[\text{PtCl}_2(\text{NH}_3)_2] \rightarrow$ Square planar

B. $[\text{Co}(\text{NH}_3)_6]\text{Cl}_3 \rightarrow$ Octahedral

C. $[\text{NiCl}_4]^{2-} \rightarrow$ Tetrahedral

D. $[\text{Fe}(\text{CO})_5] \rightarrow$ Trigonal bipyramidal

Q83 Text Solution:

For a **zero-order reaction**:

$$[R] = [R_0] - kt$$

So the plot of $([R])$ vs time is a **straight line with negative slope**.

$$\text{slope} = -k$$

Hence, the reaction is **zero order**.

Q84 Text Solution:

SCN^- is an **ambidentate ligand** because it can coordinate through either:

S

or

N

So it forms linkage isomers: thiocyanato-S and isothiocyanato-N.

Q85 Text Solution:

$$\Delta H^\circ = \Delta U^\circ + \Delta n_g RT$$

$$\Delta n_g = 2 - (2 + 1) = -1$$

$$\Delta H^\circ = -10 - \frac{(8.31)(298)}{1000} = -12$$

$$, 476 \text{ kJ mol}^{-1}$$

$$\Delta G^\circ = \Delta H^\circ - T \Delta S^\circ$$

$$= -12, 476 -$$

$$298 \times (-44)/1000$$

$$= +0. 6356 \text{ kJ mol}^{-1}$$

Since $\Delta G^\circ > 0$, reaction is **non-spontaneous**.



Q86 Text Solution:

For the given ozone structure:

- O atom 2 is double bonded $\rightarrow$ formal charge (0)
- Central O atom 1 has one double bond and one single bond $\rightarrow$ formal charge +1
- O atom 3 is single bonded with three lone pairs $\rightarrow$ formal charge -1

So, for atoms 2, 1, 3 respectively:

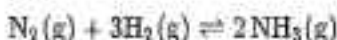
0, +1, -1

Q87 Text Solution:

$$K_p = K_c(RT)^{\Delta n}$$

For $K_p < K_c$, Δn should be negative.

For:



$$\Delta n = 2 - (1 + 3) = -2$$

So,

$$\Delta n = 2 - (1 + 3) = -2$$

Q88 Text Solution:

Arrhenius equation:

$$\ln k = \ln A - \frac{E_a}{RT}$$

Given:

$$\ln k = 14.34 - \frac{1.25 \times 10^4}{T}$$

So,

$$\frac{E_a}{R} = 1.25 \times 10^4$$

$$E_a = 1.25 \times 10^4 \times 1.987 = 24837.5$$

$$\text{cal mol}^{-1}$$

$$= 24.84 \text{ kcal mol}^{-1}$$

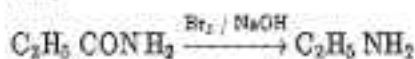
Q89 Text Solution:

Reason:

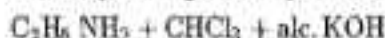


AgCN gives **isocyanide**, which has foul smell.

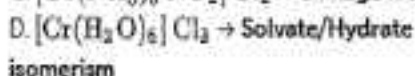
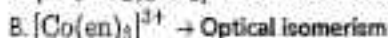
Also,



Then ethylamine gives carbylamine reaction:



So (Z) is **ethyl isocyanide**.

Q90 Text Solution:**Q91 Text Solution:**

The **Evil Quartet** of biodiversity loss includes:

1. **Habitat loss and fragmentation**
2. **Over-exploitation**
3. **Alien species invasions**
4. **Co-extinctions**

Q92 Text Solution:

Nucleolus is the site of **active rRNA synthesis** and ribosomal subunit assembly.

So, for ribosomal RNA synthesis $\rightarrow$ **Nucleolus**.

Q93 Text Solution:

- **G₁ phase** $\rightarrow$ Cell grows, metabolically active, DNA not replicated $\rightarrow$ I
- **S phase** $\rightarrow$ DNA synthesis, DNA amount doubles $\rightarrow$ III
- **G₂ phase** $\rightarrow$ Proteins synthesized, growth continues $\rightarrow$ IV
- **M phase** $\rightarrow$ Actual cell division occurs $\rightarrow$ I

Q94 Text Solution:

- **Productivity** $\rightarrow$ Rate of biomass production $\rightarrow$ III
- **Net primary productivity (NPP)** $\rightarrow$ GPP - respiration losses $\rightarrow$ I
- **Gross primary productivity (GPP)** $\rightarrow$ Rate of production of organic matter during photosynthesis $\rightarrow$ IV
- **Secondary productivity** $\rightarrow$ Rate of formation of new organic matter by consumers $\rightarrow$ II

Q95 Text Solution:

- **A correct:** Amazon rainforest cleared for soya beans = **habitat loss**.
- **B correct:** Steller's sea cow and passenger pigeon became extinct due to **over-exploitation**.



- **C incorrect:** Nile perch caused extinction/decline of many **cichlid fishes**, not growth.
- **D correct:** Water hyacinth is an **invasive alien species**.
- **E incorrect:** In co-extinction, associated plant/animal species **are affected** and may also become extinct.

Q96 Text Solution:

- **A incorrect:** Lipids are generally **water-insoluble**.
- **B correct:** Proteins are **polypeptides**.
- **C correct:** Polysaccharides are **long chains of sugars**.
- **D incorrect:** Adenine and guanine are **purines**, not substituted pyrimidines.
- **E correct:** Almost all enzymes are **proteins**.

Q97 Text Solution:

For **1 glucose molecule**, Calvin cycle fixes **6 CO₂**.

Requirement per CO₂ = **3 ATP + 2 NADPH**

So for **6 CO₂**:

$$\text{ATP} = 6 \times 3 = 18$$

$$\text{NADPH} = 6 \times 2 = 12$$

Hence, **18 ATP and 12 NADPH**.

Q98 Text Solution:

- **A true:** Restriction endonucleases are called **molecular scissors**.
- **B true:** They restrict growth of bacteriophages in bacteria like *E. coli*.
- **C not true:** They do **not always cut only at the centre** of palindromic sites; cuts may be staggered or blunt.
- **D not true:** Removing nucleotides from DNA ends is done by **exonucleases**, not endonucleases.
- **E true:** They recognise specific **palindromic sequences**.

Q99 Text Solution:

- **Decomposition** → Breaking down complex organic matter into inorganic substances → **III**

- **Detritus** → Dead remains of plants and animals including fecal matter → **IV**
- **Mineralisation** → Release of inorganic nutrients by microbes → **II**
- **Humification** → Accumulation of dark coloured amorphous colloidal substance → **I**

Q100 Text Solution:

In **gymnosperms**, ovules are **not enclosed by ovary wall**, so seeds remain exposed/naked.

Pinus is a gymnosperm.

Q101 Text Solution:

- **Marginal placentation** → Pea
- **Axile placentation** → Lemon
- **Parietal placentation** → Mustard
- **Basal placentation** → Marigold

Q102 Text Solution:

Root hairs arise from the **epidermal cells** in the region of **maturation** of the root. They help in absorption of water and minerals.

Q103 Text Solution:

In the **phase of elongation**, cells show:

- **Cell enlargement**
- **Increased vacuolation**
- **New cell wall deposition**

Large conspicuous nuclei are mainly characteristic of **meristematic cells**, not elongation phase.

Q104 Text Solution:

All statements are **correct**.

A transcription unit has:

- **Promoter**
- **Structural gene**
- **Terminator**

Promoter is near the **5' end** of structural gene, binds **RNA polymerase**, and helps decide **template/coding strand**. Terminator is near the **3' end of coding strand** and ends transcription.

Q105 Text Solution:



α -helix and β -pleated sheet are forms of secondary structure of proteins.

Q106 Text Solution:

- **A correct:** Amino acids are substituted methanes.
- **B incorrect:** Serine is not aromatic; it is a neutral amino acid.
- **C correct:** Valine is a neutral amino acid.
- **D incorrect:** Lysine is basic, not acidic.

Q107 Text Solution:

In grasses, **bulliform cells** become flaccid during water stress and make the leaves **curl inward**, reducing water loss.

Q108 Text Solution:

Incorrect statements:

- **A incorrect:** Water-splitting complex is associated with PS II, not PS I.
- **D incorrect:** C_4 plants exhibit Kranz anatomy, not C_3 plants.

Correct statements:

- **B correct:** C_4 plants ultimately use the C_3 /Calvin pathway as main biosynthetic pathway.
- **C correct:** Photorespiration is essentially absent/negligible in C_4 plants.
- **E correct:** ATP synthesis in chloroplast occurs by chemiosmosis.

Q109 Text Solution:

- **Conjunctive tissue** $\rightarrow$ Tissue between xylem and phloem $\rightarrow$ III
- **Casparian strips** $\rightarrow$ Endodermal cells with suberin deposition $\rightarrow$ IV
- **Subsidiary cells** $\rightarrow$ Specialised cells near guard cells $\rightarrow$ I
- **Starch sheath** $\rightarrow$ Endodermal cells rich in starch $\rightarrow$ II

Q110 Text Solution:

- **Genetically modified organism** $\rightarrow$ Bt cotton $\rightarrow$ II

- **Thermostable DNA polymerase** $\rightarrow$ *Thermus aquaticus* $\rightarrow$ III
- **Ti plasmid** $\rightarrow$ *Agrobacterium tumefaciens* $\rightarrow$ I
- **pBR322** $\rightarrow$ cloning vector of *Escherichia coli* $\rightarrow$ IV

Q111 Text Solution:

Heterophylly means different leaf forms in response to environment. This ability of plants to alter development according to environmental conditions is called **plasticity**.

Q112 Text Solution:

In **racemose inflorescence**, main axis keeps growing and flowers are arranged **acropetally** — older flowers at the base, younger flowers towards the apex.

Q113 Text Solution:

In sickle-cell anaemia, **glutamic acid is replaced** by valine at the **6th position of the β -globin chain** of haemoglobin.

Q114 Text Solution:

- **Incomplete dominance** $\rightarrow$ Flower colour in *Antirrhinum* $\rightarrow$ II
- **Co-dominance** $\rightarrow$ ABO blood groups $\rightarrow$ IV
- **Pleiotropy** $\rightarrow$ Phenylketonuria in humans $\rightarrow$ III
- **Polygenic inheritance** $\rightarrow$ Human skin colour $\rightarrow$ I

Q115 Text Solution:

Correct sequence of microsporogenesis:

Sporogenous tissue $\rightarrow$ **Pollen mother cells** $\rightarrow$ **Microspore tetrads** $\rightarrow$ **Pollen grains**

Q116 Text Solution:

Correct DNA fingerprinting sequence:

A Isolation of DNA and digestion by restriction endonucleases
 $\rightarrow$ **E** Separation of DNA fragments by electrophoresis
 $\rightarrow$ **C** Transfer to synthetic membrane
 $\rightarrow$ **B** Hybridisation with labelled VNTR probe
 $\rightarrow$ **D** Detection by autoradiography

Q117 Text Solution:



NCERT defines **bioprospecting** as exploring molecular, genetic and species-level diversity for products of economic importance.

Q118 Text Solution:

- **A true:** Fertilised egg → female honeybee (queen/worker).
- **B true:** Unfertilised egg → male by parthenogenesis.
- **C true:** Male is haploid, female is diploid.
- **D false:** Males produce sperms by mitosis, not meiosis.
- **E true:** Honeybee shows haplodiploid sex determination.

Q119 Text Solution:

1. **Denaturation:** DNA strands separate
2. **Annealing:** Primers bind
3. **Extension:** Taq polymerase extends primers

Q120 Text Solution:

- **A correct:** DNA is cut by restriction enzymes, called molecular scissors.
- **B correct:** DNA fragments separate by size in agarose gel electrophoresis.
- **C incorrect:** DNA fragments are not visible without staining.
- **D incorrect:** Ethidium bromide-stained DNA is seen under UV light, not visible light.

Q121 Text Solution:

NCERT: Whittaker's Five Kingdom classification criteria include cell structure, body organisation, mode of nutrition, reproduction, and phylogenetic relationships. Presence of flagellum is not a main criterion. So, C is excluded.

Q122 Text Solution:

After triple fusion, the central cell becomes the primary endosperm cell (PEC), which is triploid (3n).
Zygote is 2n, synergid is n.

Q123 Text Solution:

- A ☒ Histone octamer = 8 histone molecules
- B ☒ Histones are positively charged basic proteins, not negatively charged.
- C ☒ Rich in lysine and arginine
- D ☒ Negatively charged DNA wraps around positively charged histone octamer.
- E ☒ Higher packaging needs Non-histone Chromosomal (NHC) proteins.

Q124 Text Solution:

In situ conservation = conservation in the natural habitat.

NCERT lists sacred groves under in situ conservation.

Wildlife safari parks, botanical gardens, and seed banks are ex situ methods.

Q125 Text Solution:

In lac operon:

z gene → β -galactosidase, y gene → permease, a gene → transacetylase, and i gene → repressor.

Q126 Text Solution:

- A. 2,4-D → Herbicide
- B. GA₃ → Brewing industry
- C. Kinetin → Nutrient mobilisation
- D. ABA → Stimulation of stomatal closure

Q127 Text Solution:

Correct sequence:

Isolation of single cells → Digestion of cell walls → Isolation of naked protoplasts → Fusion of protoplasts → Growing hybrid protoplast into new plant

So, D → A → B → C → E.

Q128 Text Solution:

$RQ = \text{CO}_2 \text{ evolved} / \text{O}_2 \text{ consumed}$

$= 102 / 145 = 0.70 \text{ approx.}$

So RQ is between 0.5 and 0.95. This indicates a fat/lipid-like substrate (Tripalmitin).

Q129 Text Solution:

The ongoing sixth extinction is different because its rate is 100–1000 times faster than pre-human times, mainly due to human activities.

Q130 Text Solution:



- A. Trypsin → Enzyme
 B. Morphine → Alkaloid
 C. Concavealin A → Lectin
 D. Collagen → Intercellular ground substance

Q131 Text Solution:

Statement (3) is not true because the first word represents genus, and the second word represents specific epithet.

Example: *Mangifera indica* → *Mangifera* = genus, *indica* = specific epithet.

Q132 Text Solution:

In the Calvin cycle, CO_2 fixation/carboxylation is catalysed by RuBisCO .

Q133 Text Solution:

Floral formula of Solanaceae is:



It has **actinomorphic**, bisexual flower; **united sepals**, **united petals**, **5 epipetalous stamens**, **bicarpellary syncarpous superior ovary**.

Q134 Text Solution:

Xenogamy = transfer of pollen grains from anther to stigma of a different plant of the same species. So it brings genetically different pollen grains to the stigma.

Q135 Text Solution:

- A. Glycolysis → Cytoplasm
 B. ETS → Inner mitochondrial membrane
 C. Accumulation of protons → Intermembrane space
 D. Krebs' cycle → Mitochondrial matrix

Q136 Text Solution:

In pBR322, the *Bam*HI site lies within the **tetracycline resistance gene (tet^R)**. So insertion of foreign DNA at *Bam*HI causes **insertional inactivation of tet^R** , leading to loss of **tetracycline resistance**.

Q137 Text Solution:

In sickle-cell anaemia, the 6th codon of β -globin changes from GAG → GUG, causing Glu → Val substitution and Hb polymerisation.

Q138 Text Solution:

Correct option: (2)

GIFT, which stands for **Gamete Intrafallopian Transfer**, is a specific assisted reproductive technique designed for women who cannot produce their own eggs but possess a healthy reproductive environment.

Based on the definition provided:

- **The Procedure:** It involves the transfer of an **ovum (egg)** collected from a donor directly into the **fallopian tube** of another female.
- **The Recipient:** This technique is used for a female who cannot produce her own ovum but can provide a **suitable environment** for fertilization and subsequent development.

Unlike standard IVF where fertilization happens in a laboratory, in GIFT, the fertilization takes place **naturally** inside the recipient's fallopian tube.

Option (4) is ZIFT, not GIFT.

Q139 Text Solution:

Ophrys orchid shows **sexual deceit** by mimicking the female bee/wasp, attracting the male insect for **pseudocopulation**, which helps in pollination.

Q140 Text Solution:

Correct sequence of appearance:

Ramapithecus → *Homo habilis* → *Homo erectus* → *Neanderthal* → *Homo sapiens*

This follows the general NCERT human evolution order from ape-like ancestor to modern human.

Q141 Text Solution:

A. The foetus movement starts and hair appears on the head → **20 weeks of pregnancy**

B. The foetus develops limbs and digits → **8 weeks of pregnancy**

C. The foetus develops external genital organs → **12 weeks of pregnancy**

D. The foetus body is covered with fine hair; eyelids separate and eyelashes are formed → **24 weeks of pregnancy**

Q142 Text Solution:



These features match Cyclostomata:
ectoparasitic on fishes, circular sucking mouth,
6–15 pairs of gill slits, no scales and no paired
fins.

Example: *Petromyzon* (Lamprey).

Q143 Text Solution:

- A ☐ Spermatogonia divide via mitosis
(equational division) to maintain their population,
not directly via meiosis.
B ☐ Primary spermatocytes undergo meiosis I,
not mitosis.
C ☒ Secondary spermatocytes undergo meiosis
II to form haploid spermatids.
D ☐ Spermatids do not form spermatozoa by
mitosis.
E ☒ Spermatids transform into spermatozoa by
spermiogenesis.

Q144 Text Solution:

Mother heterozygous A = I^Ai

Father heterozygous B = I^Bi

Cross: I^Ai × I^Bi → offspring:

I^AI^B, I^Ai, I^Bi, ii

So O blood group (ii) probability = 1/4 = 25%.

Q145 Text Solution:

Correct order:

Fall in GFR → Renin converts Angiotensinogen
into Angiotensin I, followed by Angiotensin II →
Vasoconstriction by Angiotensin II and release of
Aldosterone → Reabsorption of Na⁺ and water
from distal parts of tubule due to Aldosterone →
Increase in blood pressure and Glomerular
filtration rate.

Q146 Text Solution:

- A. ERV → 1000–1100 mL
B. RV → 1100–1200 mL
C. IRV → 2500–3000 mL
D. TV → 500 mL

Q147 Text Solution:

- A. Progestasert → Hormone-releasing IUD
B. Multiload 375 → Copper-releasing IUD
C. Diaphragm → Rubber barrier used by females
D. Saheli → Oral contraceptive

Q148 Text Solution:

Ribosomes are non-membrane-bound
organelles present in both prokaryotic and
eukaryotic cells.

Mitochondria and lysosomes are membrane-
bound; centrosome is mainly in animal cells.

Q149 Text Solution:

The pyramid of biomass in sea is generally
inverted because the biomass of fishes far
exceeds that of phytoplankton.

Energy pyramid is always upright.

Q150 Text Solution:

Aptenodytes = Penguin, a flightless bird whose
forelimbs are modified into paddle-like flippers
for swimming.

Q151 Text Solution:

Streptokinase removes clots from blood vessels.
Statins lower blood cholesterol. Lipases are used
in detergent formulations. Cyclosporin A is an
immunosuppressive agent.

Q152 Text Solution:

A is false because mitochondria are not part of
the endomembrane system.

E is false because mitochondria are double
membrane-bound, not single membrane-bound.

Q153 Text Solution:

All three are bony fishes of class Osteichthyes:
Flying fish = *Exocoetis*, Fighting fish = *Betta*,
Angel fish = *Pterophyllum*.

Q154 Text Solution:

In grasshopper, XO type sex determination
occurs:

Male = XO = 23 chromosomes, Female = XX =
24 chromosomes.

Q155 Text Solution:

Eosinophils are about 2–3% of WBCs → 2–3% of
8000 = 160–240.
Lymphocytes are about 20–25% → 20–25% of
8000 = 1600–2000.

Q156 Text Solution:



cryIAc controls cotton bollworms, while **cryIAb** controls corn borer.

Q157 Text Solution:

Nicotine	stimulates adrenal gland to release catecholamines
Morphine	effective sedative and painkiller
Heroin	depressant; slows down body functions
Cocaine	causes euphoria and increased energy

Q158 Text Solution:

Tetany	wild contraction of muscles due to low Ca^{2+}
Arthritis	inflammation of joints
Myasthenia gravis	autoimmune disorder affecting neuromuscular junction
Muscular dystrophy	progressive degeneration of skeletal muscles

Q159 Text Solution:

Male honeybees are **haploid** and produce sperms by **mitosis**, not meiosis. This is due to haplodiploid sex determination.

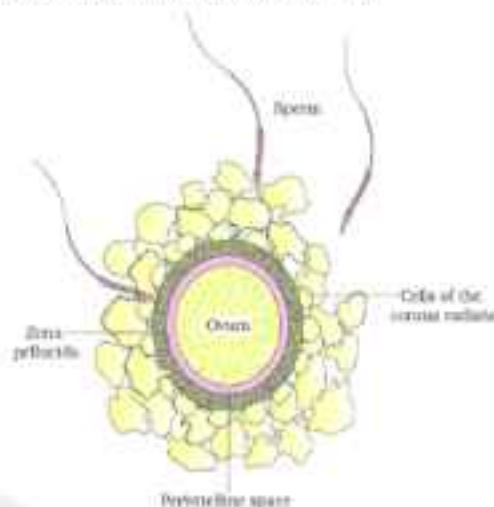
Q160 Text Solution:

1. Pulmonary ventilation (C) Air enters lungs (O_2 in) and CO_2 is expelled. This is the first step.
2. Diffusion across alveolar membrane (B) O_2 diffuses from alveoli $\rightarrow$ blood CO_2 diffuses from blood $\rightarrow$ alveoli.
3. Transport of gases by blood (E) O_2 is carried by haemoglobin to tissues CO_2 is transported back to lungs.
4. Diffusion between blood and tissues (A) O_2 moves from blood $\rightarrow$ tissues CO_2 moves from tissues $\rightarrow$ blood.
5. Cellular respiration (D) Cells use O_2 to produce energy (ATP) and release CO_2 .

Q161 Text Solution:

From outer to inner around ovum:

Corona radiata $\rightarrow$ Zona pellucida $\rightarrow$ Perivitelline space $\rightarrow$ Plasma membrane of ovum.



Q162 Text Solution:

Transgenic animals can produce useful biological products, e.g. **α -1-antitrypsin**, used to treat **emphysema**.

Q163 Text Solution:

A true: Human RBC membrane has about **52% protein**.

B true: Phospholipids are arranged in a **bilayer**.

D true: Lipid tails are **hydrophobic** and face inward.

C is false: **Mesosomes** occur in **prokaryotes**, not eukaryotic cells.

E is false: **Glycocalyx** is characteristic of **bacterial cell envelope**, not eukaryotic plasma membrane in NCERT context.

Q164 Text Solution:

Male frogs have **vocal sacs** and a **copulatory pad** on the first digit of forelimbs.

Bulging eyes, webbed feet, and olive-green skin are common frog features, not male-specific.

Q165 Text Solution:

$$\frac{dN}{dt} = rN \left(\frac{K-N}{K} \right)$$

Reason:

Verhulst-Pearl logistic growth equation includes **carrying capacity (K)** and shows growth slowing as population size **N** approaches **K**.



Q166 Text Solution:

- A true: Hepatic portal system connects intestine and liver.
C true: In female frog, ureters and oviducts open separately into cloaca.
E true: Sinus venosus opens into right atrium.
B false: Frog has 10 pairs of cranial nerves, not 12.
D false: Optic lobes are midbrain, not hindbrain.

Q167 Text Solution:

- A is incorrect: Erythroblastosis foetalis occurs when mother is Rh⁺ and foetus is Rh⁻.
E is incorrect: Anti-Rh antibodies are given immediately after the first delivery, not second.

Q168 Text Solution:

- A is false: Human skull is **dicondylic**, not monocondylic.
D is false: There are 12 pairs of ribs. Each rib is a thin flat bone connected dorsally to the vertebral column and ventrally to the sternum. It has two articulation surfaces on its dorsal end and is hence called bicephalic.
B true: Joints between adjacent vertebrae are **cartilaginous joints**.
C true: Humans have 7 cervical vertebrae.
E true: **Occipital condyles** articulate with the atlas vertebra.

Q169 Text Solution:

- Cortisol → anti-inflammatory reactions.
Aldosterone → Na⁺ and water reabsorption from renal tubules.
Cholecystokinin → pancreatic enzymes and bile juice secretion.
Progesterone → formation of alveoli in mammary glands.

Q170 Text Solution:

- Female *Anopheles* injects **sporozoites** → they reach liver → multiply asexually in liver cells → enter and multiply in RBCs → some develop into gametocytes.

Q171 Text Solution:

- Adult echinoderms are **radially symmetrical**; larvae are bilateral.

Reptiles are **poikilothermic/cold-blooded**, so they do not maintain constant body temperature.

Q172 Text Solution:

Neurotransmitters released from the pre-synaptic knob bind to **specific receptors on the post-synaptic membrane** to transmit the impulse.

Q173 Text Solution:

C is false because Ca²⁺ **activates actin sites** by removing masking, it does not inactivate actin. A, B, D and E correctly describe excitation and sliding filament mechanism.

Q174 Text Solution:

Whale and bat forelimbs are **homologous organs** with common origin but different functions, so they show **divergent evolution**, not convergent evolution.

Q175 Text Solution:

JGA is formed by cellular modifications in the DCT and the **afferent arteriole** at their contact point.

Q176 Text Solution:

Lyases catalyse removal of groups from substrates by mechanisms other than hydrolysis, often forming a **double bond (C=C)**.

Q177 Text Solution:

Molluscs → branchial respiration.
Reptiles → pulmonary respiration only.
Adult amphibians → pulmonary + cutaneous respiration.
Amoeba → cellular respiration.

Q178 Text Solution:

Large holes in Swiss cheese are due to CO₂ production by *Propionibacterium sharmanii* during fermentation.

Q179 Text Solution:

About 65 mya dinosaurs disappeared.
About 500 mya invertebrates became active.
About 350 mya jawless fishes evolved.
About 320 mya seaweeds and a few plants existed.



Q180 Text Solution:

Mutualism benefits both species.

In parasitism, parasite benefits and host is harmed.

In amensalism, one species is harmed and the other is unaffected.

A is false because in **commensalism**, one benefits and the other is **unaffected**, not harmed.



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